

DO AI WEATHER MODELS MISS EXTREMES? EVIDENCE FROM TEN MONTHS OF VERIFICATION AGAINST EUROPEAN WEATHER STATIONS

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ABSTRACT

First-generation AI weather models are often reported to underperform at extremes, mostly in reanalysis-based evaluations of deterministic regression systems. We verify twelve physical and AI forecast systems against European synoptic, solar and rain-gauge stations over ten months for 10 m wind, 2 m temperature, hourly shortwave accumulation and hourly precipitation, scoring MAE against ECMWF IFS in ERA5 1991–2020 climatological regimes. Among these systems, AI models do not show a uniform relative-skill deficit in the tails. Jua EPT-2.1 Europa leads all-conditions wind (+8.4%), while Jua EPT-2 HRRR leads temperature overall (+12.1%) and in the heat regime (+19.6 ± 2.2%). EPT-2.1 Europa and DWD ICON Global lead at gale-force wind. Jua EPT-2.1 Helios leads solar overall (+10.2 ± 1.7%), in overcast conditions (+16.4 ± 3.4%), and in the clear-sky tail (+24.8 ± 5.4%). For precipitation, three Jua models gain 14–15% at moderate intensity and 9–11% at P75–P95; EPT-2 Reasoning remains ahead at >P95 (+1.7 ± 0.5%). Failures are model-specific: ECMWF AIFS loses 4.9 ± 2.0% in the heat tail and its CRPS-trained ensemble AIFS ENS, scored on the ensemble mean, loses 5.1 ± 1.7% there, while NOAA GFS loses 22.8 ± 2.0%. Every model, NWP included, shows a shared conditional bias toward the centre of the observed distribution, with an inter-model spread several times smaller than the shared signal. Missing relative skill at extremes is therefore not a property of AI weather models as a class, but of particular AI and physical models.

1 INTRODUCTION

Data-driven weather models now outperform operational physical models on many standard forecast metrics (Bi et al., 2023; Lam et al., 2023; Ben Bouallègue et al., 2024). Their performance at extremes, however, remains an open concern. Models trained with mean-squared error tend towards the conditional mean, which can smooth sharp gradients and attenuate rare events (Olivetti & Messori, 2024a; Bonavita, 2024). Existing evaluations support this concern, reporting underestimated peak winds, hot and windy tails, and record-breaking events (Charlton-Perez et al., 2024; Olivetti & Messori, 2024b; Zhang et al., 2026).

The available evidence has two important limitations. First, it is dominated by first-generation deterministic regression models. Generative ensembles, probabilistically trained models, and high-resolution regional AI models have received much less attention. Second, most studies verify forecasts against reanalysis rather than direct station observations. Whether their conclusions extend to newer model classes and independent in situ measurements is therefore unclear.

Two questions are often conflated in this discussion: whether forecasts systematically attenuate extreme observations, and whether AI models are less accurate than physical models under the same conditions. The first concerns absolute forecast error; the second concerns relative skill. A model may underestimate an extreme event while still producing a smaller error than its physical counterpart.

We examine these questions using thirteen forecast models, including ECMWF IFS as the reference. Forecasts are verified over ten months against European synoptic, solar-radiation, and rain-gauge observations. The evaluation covers 10 m wind, 2 m temperature, hourly solar radiation, and hourly

precipitation at lead times up to 48 h. Models are compared both overall and within regimes defined by the observed value relative to a fixed 1991–2020 climatology.

Throughout the paper, “extreme” refers to the tails of this local climatological distribution, principally observations below the 5th or above the 95th quantile. These regimes identify conditions that are unusual for a given location, calendar day, and UTC hour; they are not necessarily record-breaking or impact-defined events. Precipitation is treated separately within the same framework by distinguishing dry hours and quantiles of the wet-hour distribution (Zhang et al., 2011; Perkins & Alexander, 2013).

The analysis reveals a common tendency among AI and physical models to overpredict low observations and underpredict high observations. Relative skill within those regimes, however, varies substantially by model and variable. This distinction is central: attenuation of extreme values does not imply a uniform AI-specific deficit relative to physical forecasting.

The study makes three contributions:

- It extends station-based evaluation of weather extremes to deterministic regression models, a CRPS-trained global ensemble, and high-resolution regional generative models.
- It separates relative MAE within climatological regimes from the conditional forecast errors observed in those regimes.
- It tests whether these patterns persist across four surface variables, forecast lead times, countries, and model classes, with additional categorical evaluation of heavy precipitation.

Section 3 describes the verification protocol.

2 RELATED WORK

AI weather models at extremes. Evidence of attenuated extremes has been reported in both case studies and systematic evaluations. For storm Ciarán, Charlton-Perez et al. (2024) found that FourCastNet, FourCastNet-v2, Pangu-Weather, and GraphCast reproduced the track and central pressure but underestimated peak winds by approximately 8 m s^{-1} . The same bias was not present when the models were initialised from ERA5, suggesting that smoothing in the training data was not its sole cause; the role of the verification target remained unresolved.

Across a broader sample, Olivetti & Messori (2024b) found Pangu-Weather and GraphCast competitive with IFS HRES for many regional extremes but weaker for hot and windy tails at longer leads. Zhang et al. (2026) similarly reported HRES ahead of GraphCast, Pangu-Weather, and FuXi for record-breaking events defined from ERA5. Bonavita (2024) associated this behaviour with an inability to reproduce sub-synoptic variability, arguing that deterministic AI forecasts can resemble statistical post-processing more closely than free-running simulation. A station-based evaluation by Ben Bouallègue et al. (2024) reached a more qualified conclusion: Pangu-Weather remained accurate for moderate extremes but smoothed the largest events. Collectively, this evidence is drawn mainly from deterministic regression models introduced between 2022 and 2024 and verified primarily against reanalysis.

Training objectives and generative models. Several studies connect attenuation to the training objective rather than to data-driven forecasting itself. Subich et al. (2025) attributed GraphCast’s loss of effective resolution to the double penalty induced by grid-point mean-squared error. Replacing this objective with an amplitude-preserving loss reduced the estimated effective scale from approximately 1,250 to 160 km and improved surface-wind extremes. In turbulent-flow experiments, Molinaro et al. (2024) found that ensembles of mean-squared-error models converged towards the mean flow, whereas score-based diffusion preserved the relevant statistics and spectra.

Generative weather models provide further evidence that tail deficits are not inevitable. GenCast outperformed the physical ECMWF ensemble on extremes verified against ERA5 (Price et al., 2025), while StormCast demonstrated kilometre-scale diffusion emulation of a regional convection-allowing model (Pathak et al., 2026). Direct station-based comparisons between these models, deterministic AI forecasts, and physical models nevertheless remain limited.

Station-based verification. WeatherReal (Jin et al., 2024) and MAUSAM (Gupta et al., 2025) provide in situ benchmarks for data-driven weather forecasting. MAUSAM reports errors against observations that are 15–45% larger than those measured against reanalysis, indicating that reanalysis-based evaluation can overstate forecast skill. We build on this line of work by evaluating climatological tail regimes directly against station observations using the StationBench verification framework (Schlueter & Wagner, 2025).

3 METHODS

3.1 MODELS

We evaluate twelve forecast models relative to the ECMWF IFS high-resolution deterministic forecast (Table 1). The model set spans physical and data-driven forecasting, global and regional domains, and deterministic and ensemble prediction.

The physical models comprise ECMWF ENS, NOAA GFS, DWD ICON Global, and DWD ICON-EU. The AI models include global regression models (ECMWF AIFS, Microsoft Aurora, and Jua EPT-2 Reasoning), global ensembles (ECMWF AIFS ENS and Jua EPT-2e), regional generative ensembles (Jua EPT-2 HRRR and EPT-2.1 Europa), and the solar-specialised regional model EPT-2.1 Helios (Lang et al., 2024a; Bodnar et al., 2025; Molinaro et al., 2025; Lang et al., 2024b; Siegenheim, 2025; Jua AI, 2026; Molinaro, 2026). Unlike deterministic AIFS, AIFS ENS is trained with a continuous ranked probability score objective and learned noise injection and is evaluated as a 51-member ensemble. The methodology underlying the regional generative models is described by Molinaro et al. (2026; 2024).

Models are evaluated only for variables with compatible outputs. Precipitation is available from the physical models and from EPT-2 Reasoning, EPT-2e, EPT-2 HRRR, and EPT-2.1 Europa. AIFS ENS is restricted to 10 m wind and 2 m temperature because its radiation and precipitation outputs cannot be matched to the hourly accumulations used for verification.

3.2 STATION OBSERVATIONS AND EVALUATION

Station data. Forecasts are verified against three complementary in situ networks over Europe:

- **Synoptic network:** quality-controlled observations of 10 m wind and 2 m temperature.
- **Solar-radiation network:** hourly global horizontal irradiance (GHI) from DWD, Météo-France, MeteoSwiss, KNMI, RMI, DMI, MET Norway (Frost), GeoSphere Austria, SMHI, FMI, BSRN, and ARPA Lombardia.
- **Rain-gauge network:** hourly precipitation from 4,005 gauges operated by DWD, Météo-France, MET Norway, AEMET, GeoSphere Austria, the UK Met Office, KNMI, RMI, CHMI, DMI, and MeteoSwiss. A separate rain-gauge network is required because the synoptic feed does not provide usable hourly precipitation accumulations.

The evaluation domain comprises thirteen countries: DE, FR, GB, ES, IT, PL, NL, BE, CH, NO, AT, CZ, and DK. Precipitation coverage excludes Italy and Poland. Gridded forecasts are bilinearly interpolated in latitude and longitude to the observation locations (Molinaro et al., 2026). Precipitation is scored only where gauges fall within a model’s forecast domain; regional models are compared using a common gauge sample within each evaluation cell.

None of the Jua models was trained or fine-tuned on these observations. Historical station errors are used only for the causal bias correction described in Section 3.4.

Evaluation tracks. Differences in model availability and output cadence motivate two complementary evaluation tracks:

- **Track A** covers 1 September 2025 to 30 June 2026. Wind and temperature are evaluated for IFS and the ten comparator models available throughout this period, with ICON-EU reserved for Track B. The common lead grid runs from 6 to 48 h at 6-hourly intervals, matching the output cadence of AIFS, AIFS ENS, and Aurora.

Table 1: Forecast models included in the evaluation. ECMWF IFS is the reference model.

Model	Model class	Forecast type	Domain
ECMWF IFS (reference)	Physical	Deterministic	Global
ECMWF ENS	Physical	Ensemble	Global
NOAA GFS	Physical	Deterministic	Global
DWD ICON Global	Physical	Deterministic	Global
DWD ICON-EU	Physical	Deterministic	Regional
ECMWF AIFS	Regression AI	Deterministic	Global
Microsoft Aurora	Regression AI	Deterministic	Global
Jua EPT-2 Reasoning	Regression AI	Deterministic	Global
Jua EPT-2e	Regression AI	Ensemble	Global
Jua EPT-2.1 Helios	Regression AI (solar)	Deterministic	Regional
ECMWF AIFS ENS	Probabilistic AI	Ensemble	Global
Jua EPT-2 HRRR	Generative AI	Ensemble	Regional
Jua EPT-2.1 Europa	Generative AI	Ensemble	Regional

Solar radiation is evaluated over the same period from 1 to 48 h at hourly intervals. EPT-2.1 Helios is included, whereas AIFS, AIFS ENS, Aurora, and ECMWF ENS are omitted because they lack compatible hourly solar output. The solar network has no stations in GB, ES, or PL. Italian coverage, provided mainly by ARPA Lombardia, ends in February 2026, so solar results from March onwards exclude Italy.

Precipitation is evaluated over the same 1 to 48 h range for models with compatible output. Models with hourly forecasts contribute to the complete lead grid. ECMWF ENS contributes only at its native 6-hourly leads and should therefore be interpreted as a lower-cadence comparator in aggregate results.

- **Track B** covers 1 March to 30 June 2026 and compares EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU over a common period and hourly lead grid. Wind, temperature, solar radiation, and precipitation are evaluated from 1 to 48 h. EPT-2.1 Helios is additionally included in the solar comparison.

3.3 OBSERVED-VALUE REGIMES

Regimes are defined relative to the fixed 1991–2020 ERA5 climatology (Hersbach et al., 2020), rather than to the evaluation period. ERA5 fields are bilinearly interpolated from the 0.25° grid to each station. For every station, calendar day, and UTC hour, values are drawn from a cyclic ± 15 -day window over the 30-year reference period. The window contains 930 climatological values before variable-specific conditioning and preserves both seasonal and diurnal variability.

The reference distribution depends on the support of the variable. Wind and temperature use all climatological values. Solar radiation and precipitation both contain a structural mass at zero and require conditioning. Solar thresholds are estimated from positive daytime GHI. For precipitation, observations below 0.1 mm form a separate dry regime, while the remaining thresholds are estimated from wet hours only.

We write $q_\alpha(s, d, h)$ for the α -quantile of the relevant reference distribution at station s , calendar day d , and UTC hour h . Wind, temperature, and solar radiation use $q_{0.05}$, $q_{0.25}$, $q_{0.75}$, and $q_{0.95}$. Precipitation uses the wet-hour quantiles $q_{0.50}^{\text{wet}}$, $q_{0.75}^{\text{wet}}$, and $q_{0.95}^{\text{wet}}$. The dependence on station, day, and hour is omitted below for readability.

Each forecast–observation pair is assigned to a regime according to its observed value y . Regime membership is therefore identical across models and lead times. The meteorological labels in Tables 2 and 3 are shorthand for the corresponding climatological intervals.

Defining extremes through exceedances of a fixed local climatology follows established practice in the climate-extremes literature (Zhang et al., 2011; Klein Tank et al., 2009; Perkins & Alexander,

Table 2: Observed-value regimes for wind, temperature, and solar radiation. For solar radiation, the quantiles refer to positive daytime GHI.

Regime	Definition	Wind	Temperature	Solar
All	–	all conditions	all conditions	all daytime
Very low	$y < q_{0.05}$	calm	severe cold	heavily overcast
Low	$q_{0.05} \leq y < q_{0.25}$	light air	cold	overcast
Typical	$q_{0.25} \leq y < q_{0.75}$	moderate	moderate	mixed sky
High	$q_{0.75} \leq y < q_{0.95}$	fresh to strong	warm	mostly sunny
Very high	$y \geq q_{0.95}$	gale force	heat	near clear sky

Table 3: Observed-value regimes for hourly precipitation. Wet-hour quantiles are conditional on ERA5 accumulations of at least 0.1 mm.

Regime	Definition	Interpretation
All	–	all hours
Dry	$y < 0.1 \text{ mm}$	dry hour
Light	$0.1 \text{ mm} \leq y < q_{0.50}^{\text{wet}}$	light rain
Moderate	$q_{0.50}^{\text{wet}} \leq y < q_{0.75}^{\text{wet}}$	moderate rain
High	$q_{0.75}^{\text{wet}} \leq y < q_{0.95}^{\text{wet}}$	heavy rain
Heavy	$y \geq q_{0.95}^{\text{wet}}$	most intense hours

2013; Perkins, 2015). The 1991–2020 reference period follows the WMO climate-normal convention. Conditioning precipitation quantiles on wet hours is consistent with ETCCDI wet-day indices such as R95p. We extend the same climatological framework to wind and solar radiation.

Because the thresholds are estimated independently of the evaluation period, regime frequencies need not match their nominal quantile probabilities. Moreover, the lower spatial variance of gridded ERA5 relative to point observations makes the wind and solar tails broader than those obtained from a station-based climatology.

For wind and temperature, the upper tail is the primary regime of interest. For solar radiation, the lower and upper tails correspond to overcast and near-clear-sky conditions. Lead-time figures focus on the overcast $q_{0.05}$ – $q_{0.25}$ regime, whereas regime-level results cover the full distribution.

3.4 BIAS CORRECTION

Station-based verification can be affected by systematic offsets that are routinely removed in operational post-processing. We therefore apply a causal rolling correction to each model before computing the reported scores.

For wind, temperature, and daytime solar radiation, the correction is additive. Let w denote a valid week, h the UTC hour of the valid time, and $\mathcal{D}_{w,h}$ the matched forecast–observation pairs at hour h during the preceding four weeks. For model m , the estimated bias is

$$\bar{b}_{m,w,h} = \frac{1}{|\mathcal{D}_{w,h}|} \sum_{(f_m, y) \in \mathcal{D}_{w,h}} (f_m - y), \quad (1)$$

and forecasts valid during week w are corrected according to

$$\tilde{f}_m(t) = f_m(t) - \bar{b}_{m,w,h(t)}. \quad (2)$$

The estimate is pooled across European stations and does not depend on initialisation or lead time. The procedure is identical for every model, although the estimated bias is model-specific. For ensemble models, the same shift is applied to every member, preserving the ensemble spread.

Precipitation retains a substantial probability mass at zero and has a strongly right-skewed distribution, making an additive correction unsuitable. Let $\mathcal{P}_w$ contain precipitation forecasts and observations from the preceding four weeks, pooled across initialisation hours and lead times. Forecasts

valid during week w are multiplied by

$$a_{m,w} = \text{clip}_{[0.33, 3]} \left(\frac{\sum_{\mathcal{P}_w} y}{\sum_{\mathcal{P}_w} f_m} \right), \quad \tilde{f}_m(t) = a_{m,w} f_m(t). \quad (3)$$

This correction adjusts systematic wet or dry amount biases while preserving zero forecasts. All precipitation results use the multiplicatively corrected forecasts.

Both corrections use only information available before the valid week. Because regimes are assigned from observations, bias correction does not alter regime membership. The correction removes the recent mean bias but does not force the conditional bias within each regime to vanish.

3.5 SKILL, LEAD TIMES, AND AGGREGATION

Let m denote a forecast model and c an evaluation cell, specified by the track, country, variable, lead time, and regime. We define the MAE skill of model m relative to ECMWF IFS as

$$S_{m,c} = 1 - \frac{\text{MAE}_{m,c}}{\text{MAE}_{\text{IFS},c}}. \quad (4)$$

Positive values indicate lower MAE than IFS. Within each evaluation cell, the numerator and denominator are computed from the same forecast–observation pairs, using the common intersection of available initialisations across the models being compared and the four daily cycles at 00, 06, 12, and 18 UTC.

For ECMWF ENS, ECMWF AIFS ENS, Jua EPT-2e, EPT-2 HRRR, and EPT-2.1 Europa, the ensemble mean is used as the point forecast. The analysis therefore assesses point-forecast accuracy rather than ensemble spread or probabilistic calibration. Mean forecast-minus-observation error is also reported within each regime.

Results are reported at 1, 6, 12, 24, and 48 h, where available, and aggregated over the full lead-time grid of each comparison. Track A wind and temperature use 6-hourly leads from 6 to 48 h. Track A solar and Track B use hourly leads from 1 to 48 h, as do precipitation models with hourly output. Lower-cadence models contribute only at their native leads; no temporal interpolation is applied.

Uncertainty is estimated using a leave-one-month-out jackknife and reported as one standard error around each point estimate. For precipitation, we also evaluate exceedance of the local $q_{0.95}^{\text{wet}}$ threshold. Forecast and observed exceedances are pooled across gauges and lead times up to 48 h to compute the probability of detection (POD), false-alarm ratio (FAR), critical success index (CSI), and frequency bias, defined as the ratio of forecast to observed event counts.

In lead-time figures, colour identifies the model. Solid lines denote the generative Jua models and the CRPS-trained AIFS ENS, dash-dot lines denote regression AI models, and dashed lines denote physical models. ECMWF IFS is shown as a solid black reference.

4 RESULTS

4.1 SKILL BY LEAD TIME

Figures 1–3 and Figure 4 report MAE skill relative to ECMWF IFS by lead time for wind, temperature, solar radiation, and precipitation. Each figure shows both all-conditions performance and the corresponding tail regime. Tables S8–S10 provide the corresponding wind, temperature, and solar values.

The leading wind and temperature models remain stable across the forecast horizon. EPT-2.1 Europa retains a consistent advantage for wind, while EPT-2 HRRR leads for temperature without a clear deterioration towards 48 h.

AIFS ENS improves substantially over deterministic AIFS for wind under all conditions, approaching IFS at longer leads. This improvement does not extend to gale-force winds, where both variants remain clearly below the reference. A similar distinction appears for temperature: their overall skill changes from negative at short leads to positive at longer leads, while their heat-regime skill remains negative throughout.

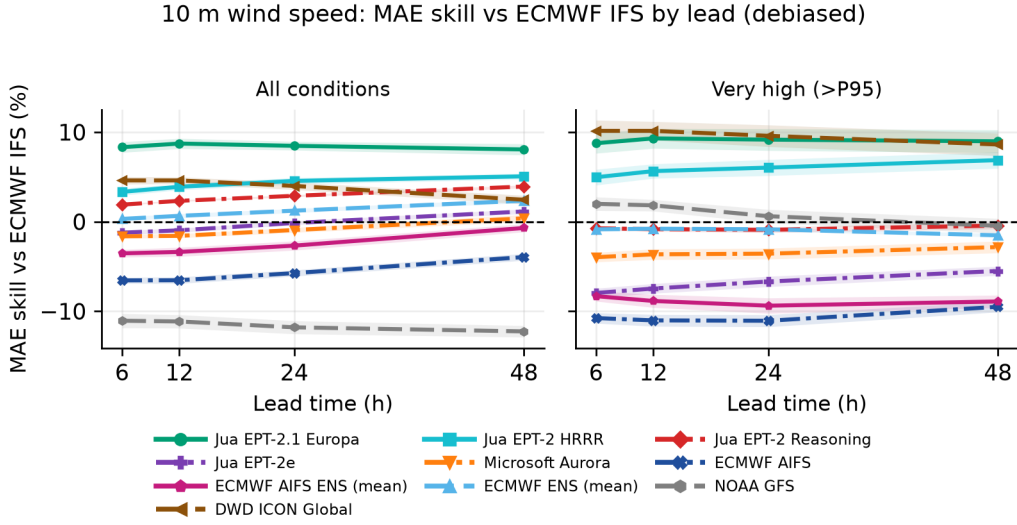


Figure 1: MAE skill relative to ECMWF IFS by lead time for 10 m wind. The left panel shows all conditions; the right panel shows the gale-force regime, $y \geq q_{0.95}$. Shading denotes ± 1 jackknife SE.

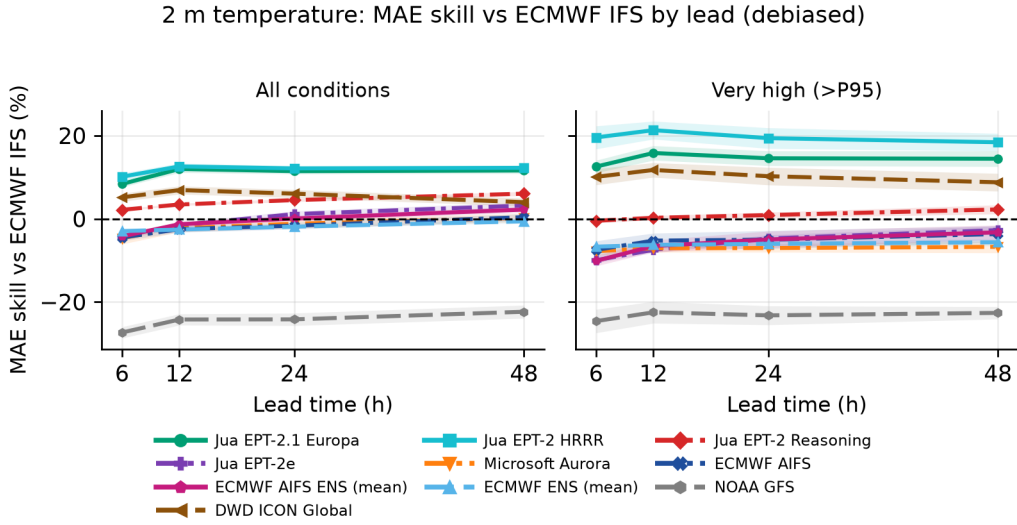


Figure 2: As Figure 1, for 2 m temperature. The right panel shows the heat regime, $y \geq q_{0.95}$.

EPT-2.1 Helios leads the solar comparison at every reported lead, with its largest advantage in the short range. Precipitation shows greater lead-time dependence: positive skill among the Jua models generally emerges after 12 h, whereas GFS and ICON Global remain at or below IFS across the reported horizon.

4.2 SKILL ACROSS THE OBSERVED DISTRIBUTION

Figures 5 and 7 report Track A MAE skill across the observed distribution. Figure 5 covers wind, temperature, and solar radiation, with values given in Tables S1–S3. Figure 7 and Table 5 provide the corresponding precipitation results. Short-range results over 1–12 h are shown in Figures 6 and 8 and Tables S4–S6 and S7. Together, these comparisons cover the full observed distribution, including very-low and dry conditions.

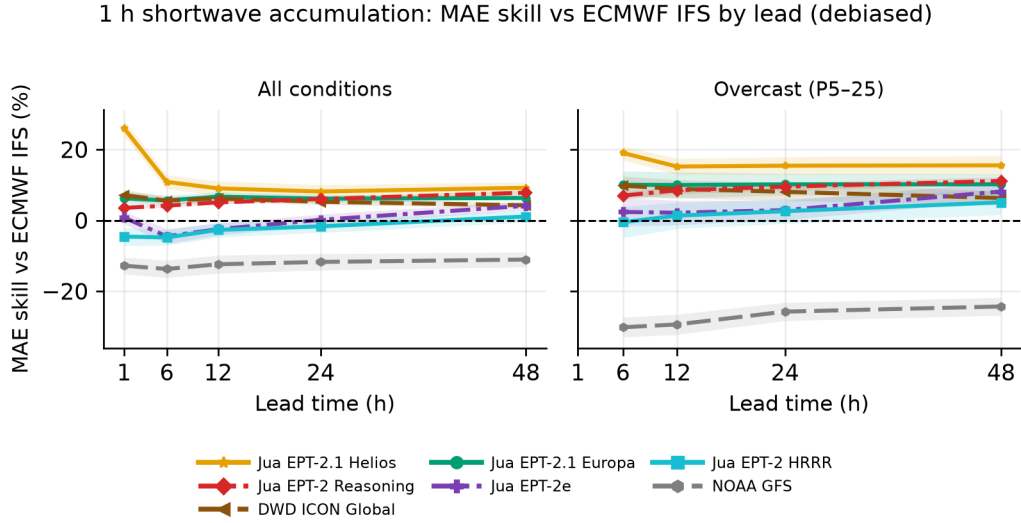


Figure 3: Track A MAE skill relative to ECMWF IFS by lead time for hourly solar radiation. The left panel shows all daytime conditions; the right panel shows the overcast regime, $q_{0.05} \leq y < q_{0.25}$. AIFS, AIFS ENS, Aurora, and ECMWF ENS are omitted because they lack compatible hourly solar output. Shading denotes ± 1 jackknife SE.

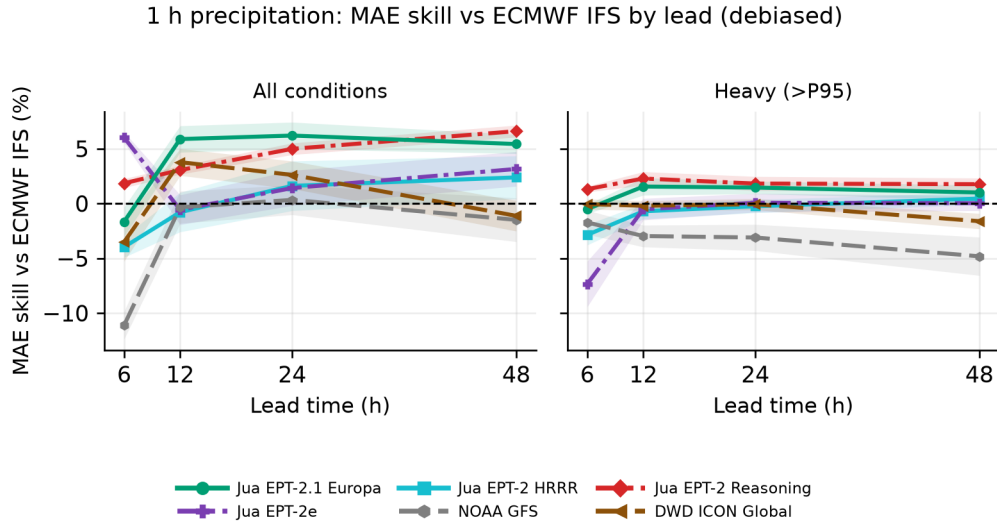


Figure 4: Track A MAE skill relative to ECMWF IFS by lead time for hourly precipitation. The left panel shows all conditions; the right panel shows the heavy regime, $y \geq q_{0.95}^{\text{wet}}$. Shading denotes ± 1 jackknife SE.

Table 4: Track A MAE skill (%) by country in the heat regime ($y \geq q_{0.95}$), aggregated over 6–48 h after bias correction. NOAA GFS is negative in every country with available observations.

Model	DE	FR	GB	ES	IT	PL	NL	BE	CH	NO	AT	CZ	DK
Jua EPT-2.1 Europa	+6.6	+16.1	-1.2	+6.4	+17.2	+6.2	-3.4	-3.0	+31.1	+5.5	+28.2	+16.6	-4.4
Jua EPT-2 HRRR	+10.5	+21.5	+4.0	+10.5	+27.6	+5.6	+1.8	-0.6	+29.4	+14.2	+16.6	+17.7	+12.2
Jua EPT-2 Reasoning	+6.6	+2.6	+4.3	-2.6	+0.1	+3.9	+3.8	-1.0	-1.3	-5.2	-3.1	+7.5	+2.7
Jua EPT-2e	+6.1	-2.2	-0.3	-15.1	-10.4	-2.4	+0.6	-2.8	-2.5	-12.2	-16.3	-1.3	+1.7
Microsoft Aurora	+2.1	-4.5	-7.5	-18.7	-7.8	-0.9	+0.7	-11.8	-3.3	-18.0	-15.4	-7.0	+1.8
ECMWF AIFS	+0.8	-1.6	-10.2	-12.7	-14.6	+1.9	-0.4	+6.3	+1.9	-3.4	-25.0	-3.1	+10.3
ECMWF AIFS ENS (mean)	+5.3	-3.1	-6.7	-10.6	-12.6	-3.1	-3.9	+5.1	+4.6	-3.2	-38.7	+2.1	+4.4
ECMWF ENS (mean)	-2.1	-4.6	-5.1	-12.1	-4.0	-8.6	+3.4	-2.9	-10.7	-9.8	-10.6	-1.5	-9.2
NOAA GFS	-18.4	-23.6	-37.0	-25.1	-13.2	-24.3	-21.9	-12.2	-10.2	-43.1	-25.3	-31.4	-24.7
DWD ICON Global	+6.9	+8.8	-4.5	+3.6	+12.8	+4.8	-2.7	+10.0	+22.7	+18.1	+16.8	+15.5	-6.5
DWD ICON-EU [†]	+13.1	+18.4	+1.3	+7.4	+20.1	+9.1	-2.6	+14.1	+34.6	+6.6	+28.1	+17.8	-0.7

For wind, EPT-2.1 Europa performs most strongly in the centre of the distribution and retains positive skill at gale force. In the upper tail, its point estimate is close to that of ICON Global, showing that the leading AI and physical models are comparably accurate in this regime. For temperature, EPT-2 HRRR has the highest overall and heat-regime estimates, followed closely by EPT-2.1 Europa; the difference between them is small relative to the jackknife uncertainty.

Tail behaviour varies substantially within both model classes. EPT-2 Reasoning performs well through the centre of the distribution but loses skill in parts of the wind and temperature tails. Aurora and EPT-2e are close to IFS overall yet deteriorate sharply in severe cold. AIFS ENS improves over deterministic AIFS in the central regimes but retains similar deficits in calm, gale-force, severe-cold, and heat conditions. The CRPS objective and ensemble averaging therefore improve the bulk of the AIFS distribution without eliminating its tail errors. NOAA GFS has the largest temperature deficit, with negative heat skill in every country reported in Table 4.

Solar radiation shows a different distributional pattern. EPT-2.1 Helios has the highest overall estimate, with its advantage concentrated in overcast and near-clear-sky conditions. In the typical irradiance regime, however, it falls below IFS and is the weakest AI model. EPT-2 Reasoning instead performs most strongly in the centre of the solar distribution. The specialised Helios model therefore improves the tails rather than the distribution uniformly. EPT-2 HRRR remains close to IFS overall and loses skill in the near-clear-sky tail.

Restricting the comparison to 1–12 h strengthens the solar advantage of EPT-2.1 Helios but leaves the principal wind and temperature conclusions unchanged. EPT-2.1 Europa remains the leading wind model, while EPT-2 HRRR and EPT-2.1 Europa retain the highest temperature estimates.

Precipitation skill is concentrated in the middle of the wet-hour distribution rather than at its extreme. EPT-2 HRRR, EPT-2e, and EPT-2 Reasoning improve on IFS by approximately 14–15% in moderate rain and retain gains of about 9–11% in the high regime. The physical global models remain below IFS across these wet regimes. EPT-2 Reasoning also has the highest all-conditions estimate.

At the heaviest precipitation intensities, the differences narrow and become model-specific. EPT-2 Reasoning remains modestly ahead of IFS, EPT-2.1 Europa is approximately neutral, and EPT-2 HRRR and EPT-2e fall slightly below the reference. The same separation is already present over 1–12 h: the Jua models retain clear gains in moderate rain but diverge in the heavy regime. Their advantage across the broader wet distribution therefore does not translate into a common advantage at its extreme.

EPT-2.1 Europa exhibits a distinct error profile. It performs strongly during dry hours but loses skill from light through moderate rain, consistent with a residual wet-amount bias after multiplicative correction. Per-country precipitation profiles are provided in Figure S4. Despite differences in relative MAE, all models underpredict the observed amount in the heaviest regime; this shared conditional-error pattern is examined in Section 4.3.

The categorical results provide a consistent picture (Figure 9 and Table S19). EPT-2.1 Europa, ICON Global, and EPT-2 Reasoning match or slightly exceed the IFS critical success index, but every model has a frequency bias below one. Heavy precipitation is therefore underdetected by both AI and physical models.

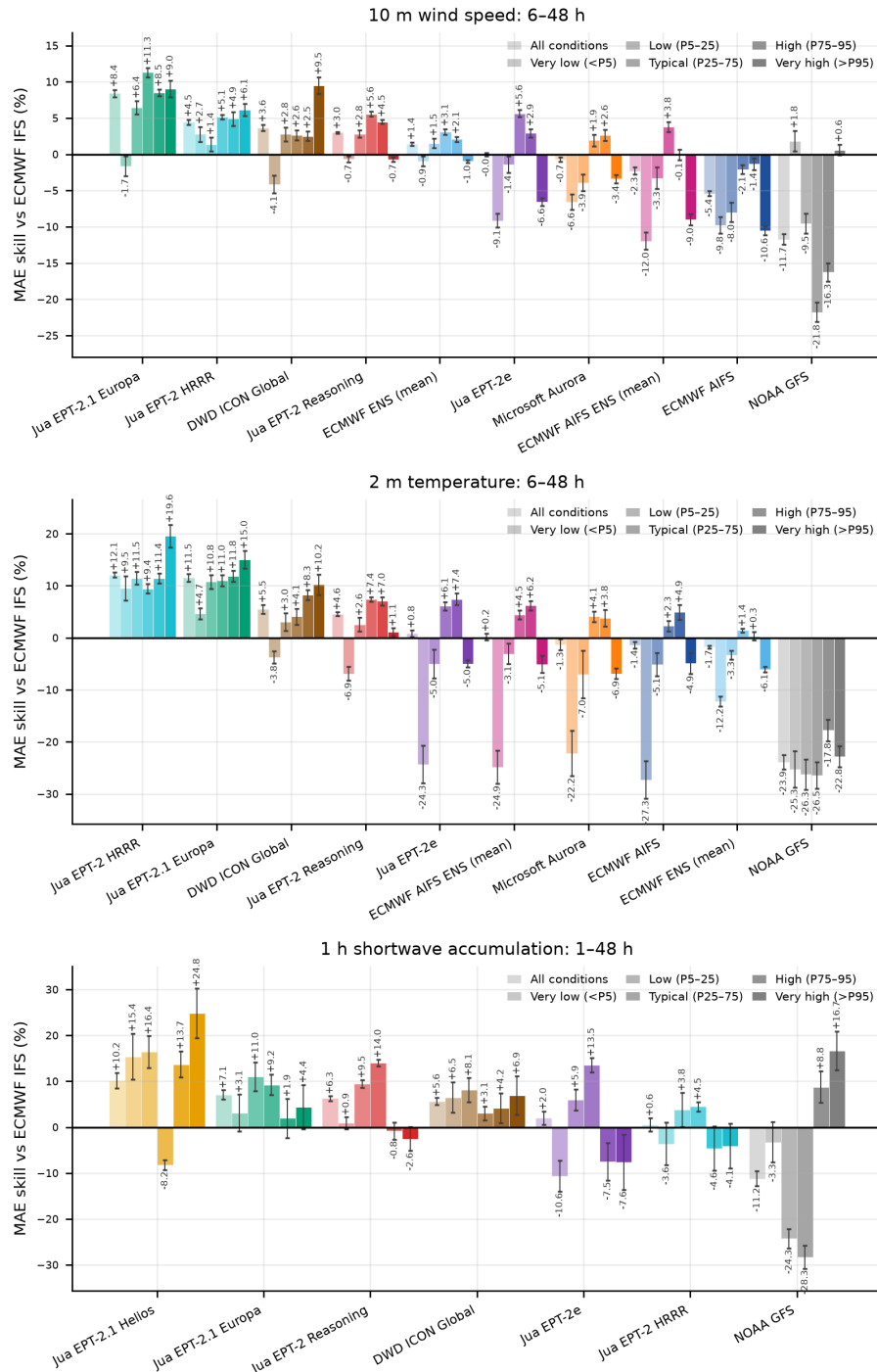


Figure 5: Track A MAE skill relative to ECMWF IFS by observed-value regime, with ± 1 jackknife SE. Wind and temperature are aggregated over 6–48 h at 6-hourly intervals; solar radiation is aggregated over 1–48 h at hourly intervals. Models are ordered by all-conditions skill within each panel.

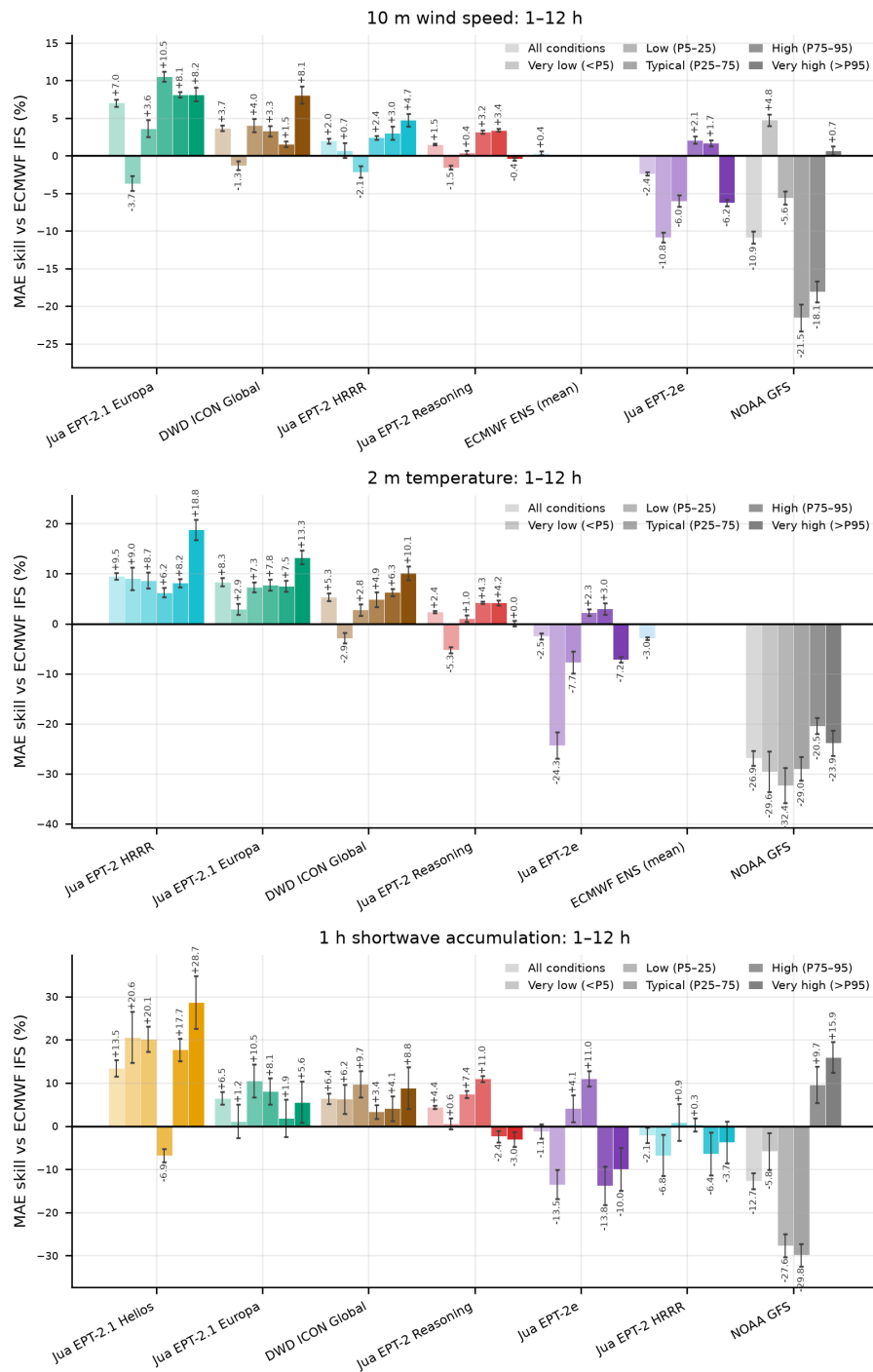


Figure 6: Track A MAE skill relative to ECMWF IFS by observed-value regime over 1–12 h, with ± 1 jackknife SE. Wind and temperature are restricted to models with hourly output; solar radiation includes EPT-2.1 Helios.

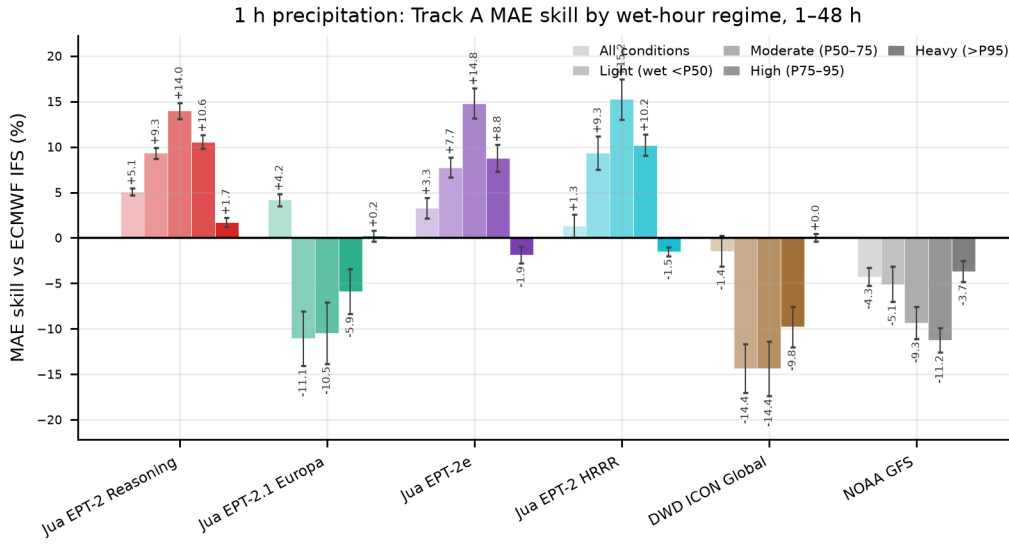


Figure 7: Track A MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h, with ± 1 jackknife SE. The dry regime is reported in Table 5. Models are ordered by all-conditions skill.

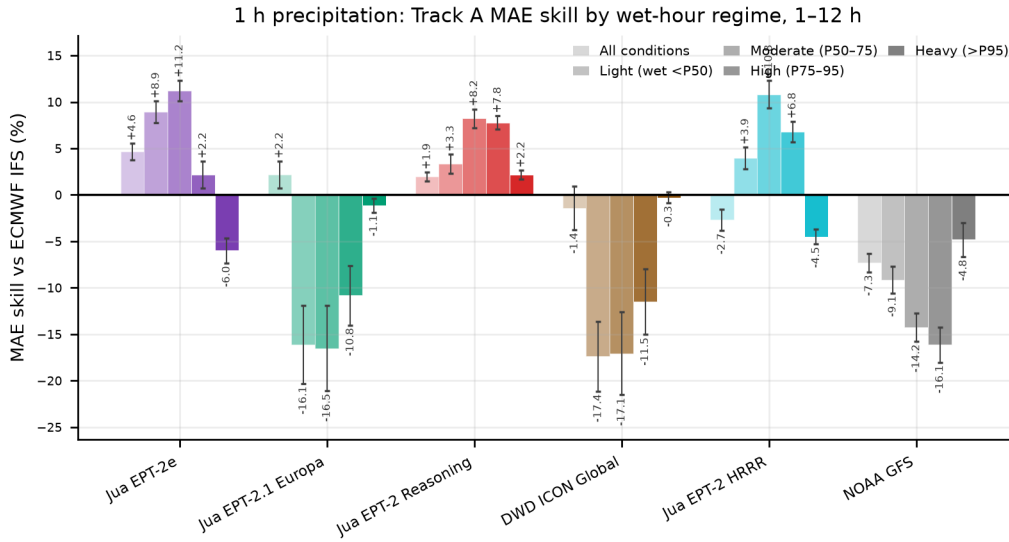


Figure 8: As Figure 7, restricted to the 1–12 h forecast range.

Table 5: Track A MAE skill (%) relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h after multiplicative bias correction. Parentheses give the leave-one-month-out jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+5.1 (± 0.4)	+2.2 (± 1.3)	+9.3 (± 0.6)	+14.0 (± 0.9)	+10.6 (± 0.8)	+1.7 (± 0.5)
Jua EPT-2 HRRR	+1.3 (± 1.3)	-6.1 (± 3.3)	+9.3 (± 1.8)	+15.2 (± 2.2)	+10.2 (± 1.2)	-1.5 (± 0.5)
Jua EPT-2e	+3.3 (± 1.1)	+1.5 (± 4.8)	+7.7 (± 1.1)	+14.8 (± 1.7)	+8.8 (± 1.5)	-1.9 (± 0.9)
Jua EPT-2.1 Europa	+4.2 (± 0.7)	+21.0 (± 2.7)	-11.1 (± 3.0)	-10.5 (± 3.4)	-5.9 (± 2.5)	+0.2 (± 0.6)
NOAA GFS	-4.3 (± 1.0)	-0.2 (± 1.8)	-5.1 (± 2.0)	-9.3 (± 1.8)	-11.2 (± 1.4)	-3.7 (± 1.2)
DWD ICON Global	-1.4 (± 1.7)	+8.3 (± 2.7)	-14.4 (± 2.7)	-14.4 (± 3.0)	-9.8 (± 2.2)	+0.0 (± 0.4)

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1 h precipitation: heavy-precip exceedance, Track A 1–48 h (debiased)

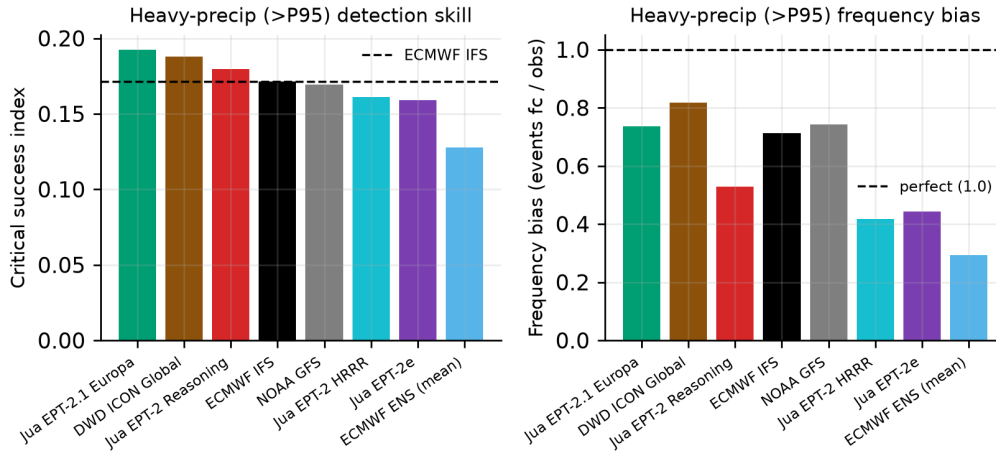


Figure 9: Detection of heavy precipitation ($y \geq q_{0.95}^{\text{wet}}$) on Track A over 1–48 h. Left: critical success index, with ECMWF IFS shown by the dashed line. Right: frequency bias, with unity denoting the correct event frequency. All models underpredict the frequency of heavy precipitation.

4.3 SHARED CONDITIONAL-ERROR PATTERN

Figure 10 reports mean forecast-minus-observation error at 48 h, conditioned on the observed-value regime. The grey envelope shows the range across all models; selected models are highlighted to make their differences visible.

Despite their different architectures and relative skill, all models show the same centre-seeking pattern: they overpredict low observations and underpredict high observations. The variation across regimes is much larger than the spread between models. The effect is strongest for wind and precipitation, with gale-force winds underpredicted by more than 2 m s^{-1} and the heaviest precipitation by approximately 2 mm. Temperature and solar radiation show the same change in error sign, although with smaller absolute differences.

Because the regimes are defined by the verifying observation, this pattern is not direct evidence of smoothing. It can also arise from limited predictability, unresolved local variability, observation and representativeness errors, or regression towards the conditional mean. The relevant finding is that this common conditional structure coexists with model-specific differences in relative MAE.

4.4 REGIONAL COMPARISON

Track B compares EPT-2.1 Europa and EPT-2 HRRR directly with ICON-EU from March to June 2026. Regime-level results over 1–48 h are shown in Figure 11, with the 1–12 h comparison in Figure 12 and lead-time curves in Figure 13. The corresponding values are reported in Tables S11–S13 and S14–S16. Precipitation results are shown in Figures 14 and 15 and Tables S18 and S17.

EPT-2.1 Europa has the highest overall wind estimate, and this ordering is maintained across lead times. At gale force, however, Europa and ICON-EU are comparable within uncertainty, while EPT-2 HRRR is weaker. The regional AI advantage is therefore clearer under all conditions than in the upper wind tail.

Temperature differences are smaller. All three regional models achieve similar overall skill and converge to approximately 17% in the heat regime. Europa has the highest full-horizon point estimate, while ICON-EU becomes more competitive over 1–12 h. Solar radiation provides a clearer separation: EPT-2.1 Helios leads overall and in both the overcast and near-clear-sky regimes, with a larger margin at short leads. Europa and ICON-EU perform similarly overall, whereas EPT-2 HRRR remains close to IFS and loses skill in the near-clear-sky tail.

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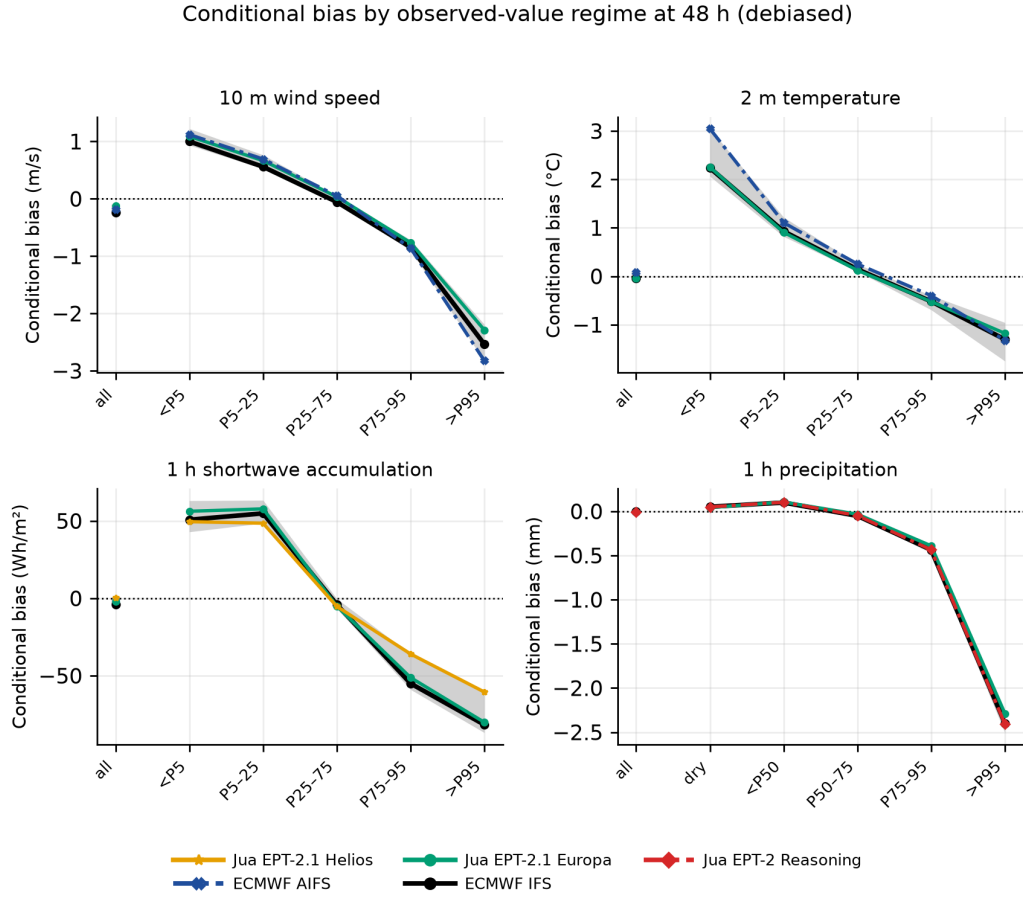


Figure 10: Mean forecast-minus-observation error at 48 h, conditioned on the observed-value regime after bias correction. The grey envelope spans all models. ECMWF IFS, EPT-2.1 Europa, and ECMWF AIFS are highlighted in every panel; the solar panel additionally highlights EPT-2.1 Helios and the precipitation panel EPT-2 Reasoning.

These results place ICON-EU alongside the leading generative models in the wind and temperature tails, while Europa retains the clearest all-conditions wind advantage. The comparison does not isolate the effect of resolution, since the models also differ in architecture, training, initialisation, and domain.

Precipitation again differs between aggregate and conditional performance. EPT-2 HRRR is the only regional model with positive skill from light through high rain, reaching its largest advantage in the moderate regime. ICON-EU is positive overall but negative across these wet regimes, while EPT-2.1 Europa repeats its Track A pattern of strong dry-hour performance and weaker skill once precipitation occurs. Aggregate skill for ICON-EU and Europa is therefore driven primarily by their treatment of dry hours.

Differences narrow in the heavy regime. Europa and ICON-EU remain close to IFS, while EPT-2 HRRR is slightly below it. Thus, HRRR performs most strongly across the broader wet distribution but does not retain that advantage at the highest intensities.

4.5 COUNTRY-LEVEL REGIME PROFILES

Figures S1–S4 report MAE skill by country and observed-value regime for March–June 2026, aggregated over hourly leads from 1 to 48 h. The figures combine the Track B regional models with selected global baselines and therefore do not constitute either the matched Track A or Track B comparison. All panels use the same ERA5 1991–2020 regime definitions introduced in Section 3.3.

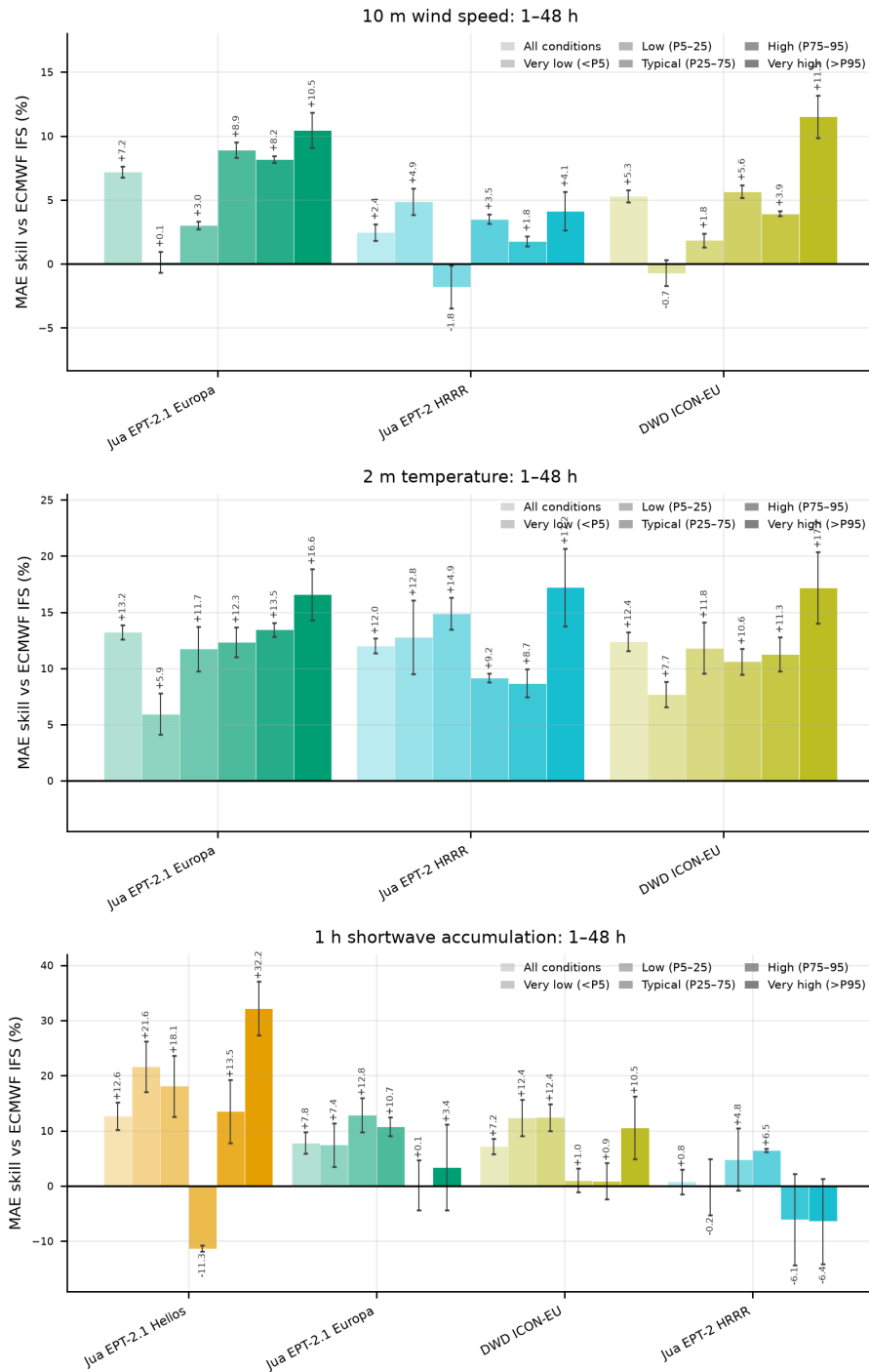


Figure 11: Track B MAE skill relative to ECMWF IFS by observed-value regime, aggregated over 1–48 h from March to June 2026, with ± 1 jackknife SE. Wind and temperature compare EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU; the solar panel additionally includes EPT-2.1 Helios. Models are ordered by all-conditions skill within each panel.

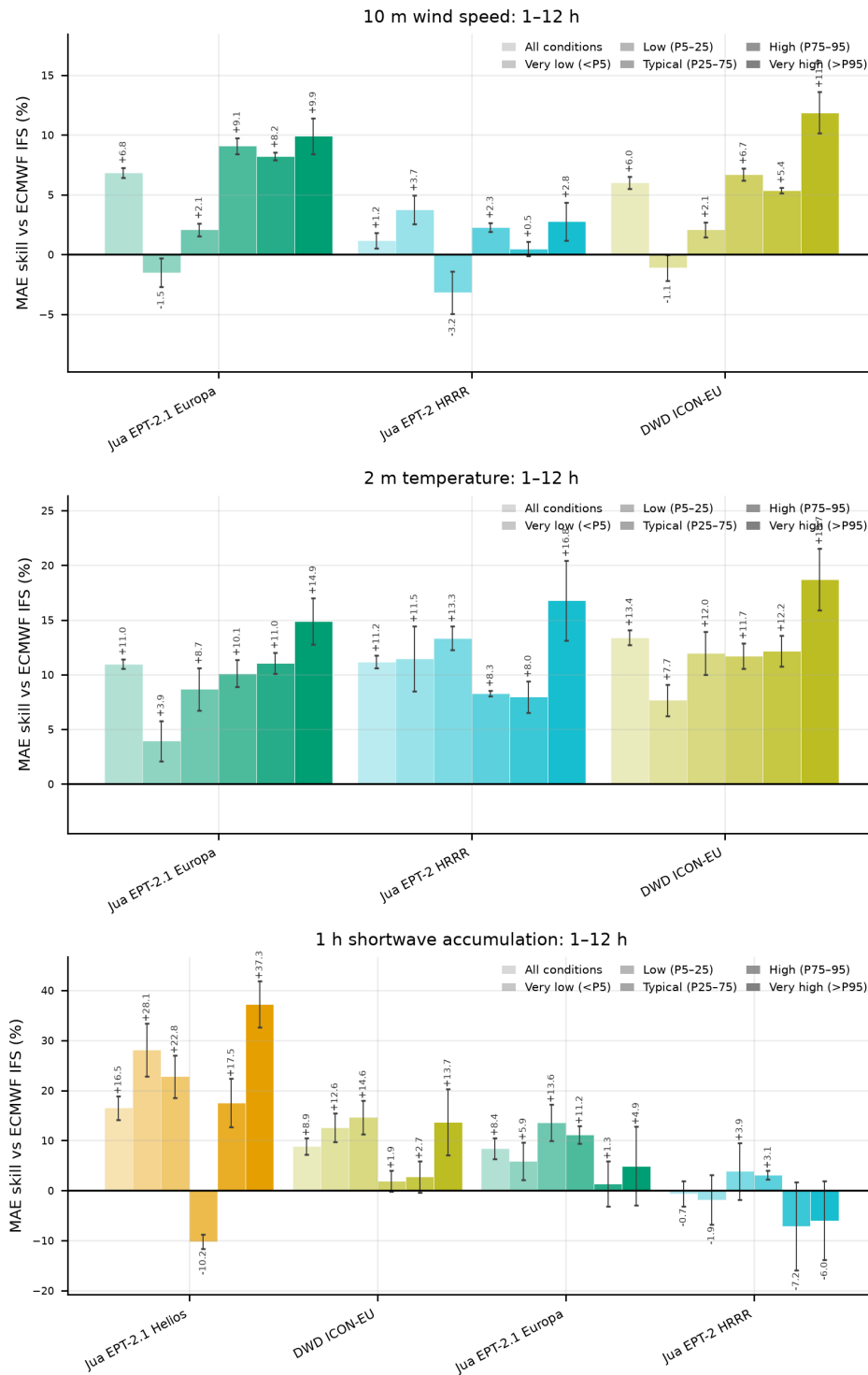


Figure 12: As Figure 11, restricted to the 1–12 h forecast range.

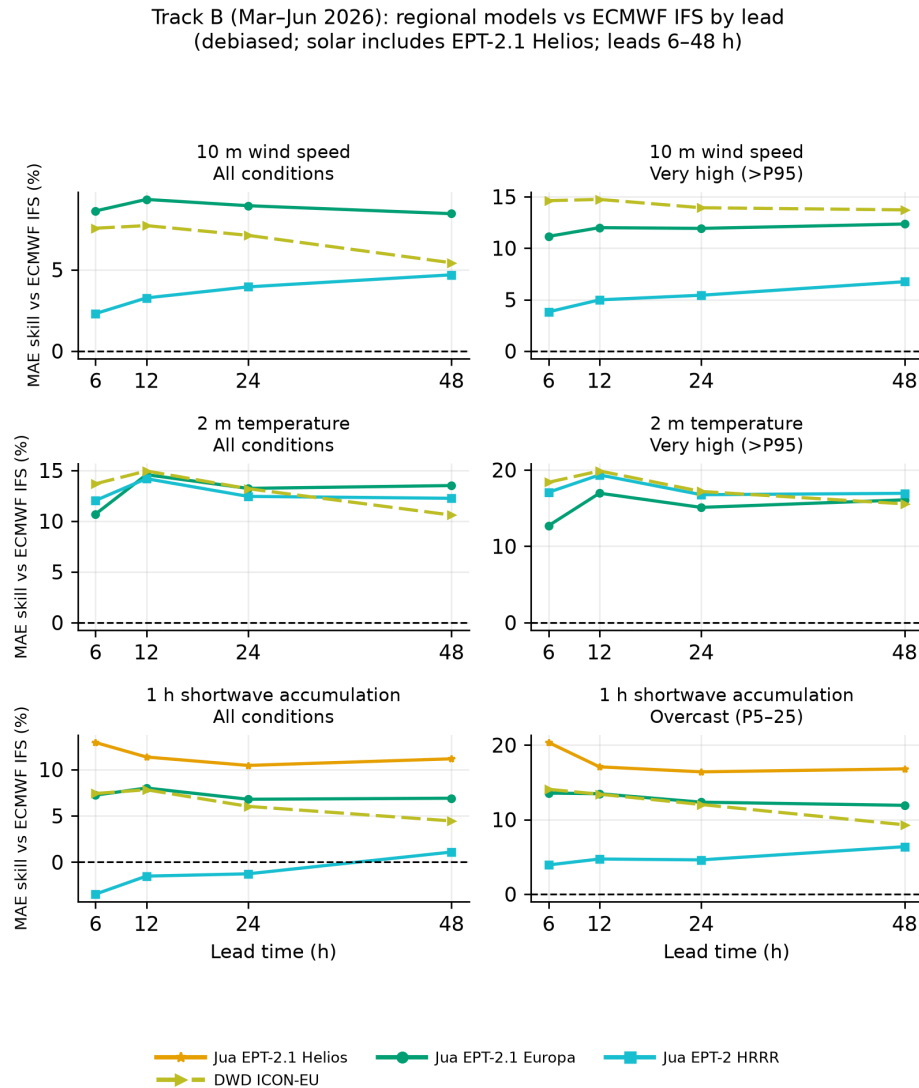


Figure 13: Track B MAE skill relative to ECMWF IFS by lead time. Rows show 10 m wind, 2 m temperature, and hourly solar radiation. Columns compare all conditions with gale-force wind, heat, and overcast conditions, respectively. Solar radiation additionally includes EPT-2.1 Helios. Plotted leads are 6, 12, 24, and 48 h; shading denotes ± 1 jackknife SE.

The country-level results support the Europe-wide patterns. EPT-2.1 Europa and ICON-EU are generally among the strongest wind and temperature models, while EPT-2.1 Helios leads the solar comparison in most countries. The temperature and heat deficits of GFS are also widespread, rather than driven by an individual country.

Countries without sufficient matched observations are omitted. The solar panels exclude GB, ES, IT, and PL, while the precipitation panels exclude IT and PL.

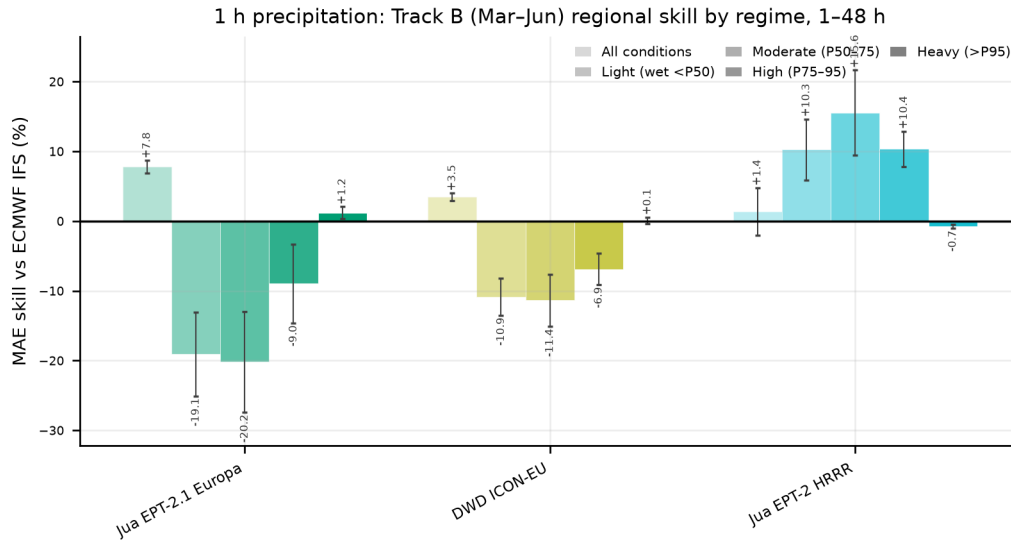


Figure 14: Track B MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h from March to June 2026, with ± 1 jackknife SE.

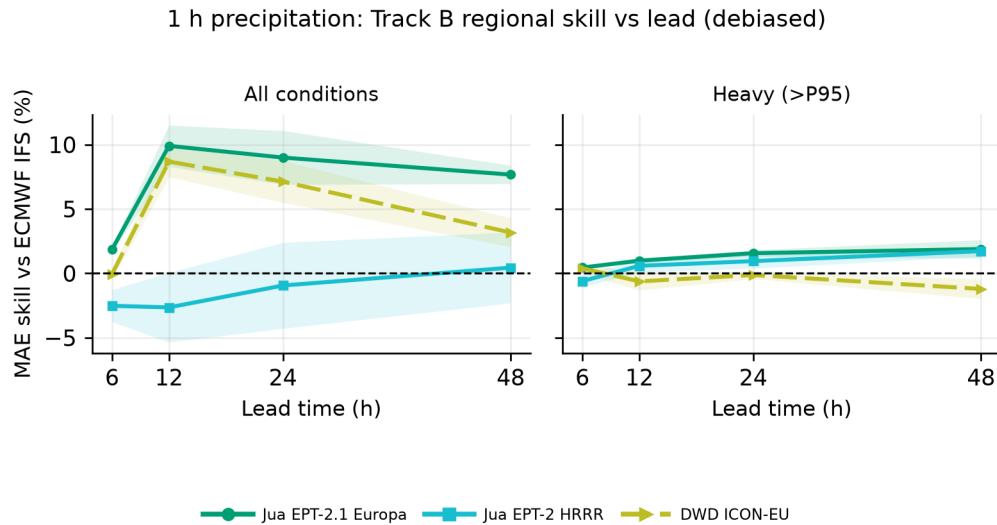


Figure 15: Track B precipitation skill relative to ECMWF IFS by lead time. The left panel shows all conditions; the right panel shows the heavy regime, $y \geq q_{0.95}^{\text{wet}}$. Shading denotes ± 1 jackknife SE.

5 DISCUSSION

The results separate two distinct meanings of “missing extremes.” First, a forecast may underestimate the magnitude of an extreme observation. Second, it may be less accurate than a competing model under the same conditions. The first behaviour is common across the evaluated models: low observations tend to be overpredicted and high observations underpredicted. The second is not. Several AI models achieve lower MAE than IFS within the heat, gale, solar, and precipitation regimes. Attenuation of extreme values therefore does not imply a general AI-specific loss of relative skill.

This distinction helps place the results in the context of earlier work. Previous studies largely considered first-generation regression models verified against reanalysis and found competitive average performance alongside underestimation of event magnitude (Olivetti & Messori, 2024b). The present station-based evaluation includes a broader range of models and reveals substantial variation across both models and variables. EPT-2.1 Helios performs strongly in the two solar tails but poorly in the typical irradiance regime. The Jua precipitation models improve most clearly through the middle of the wet distribution, while their advantage narrows at the highest intensities. Relative skill at extremes is therefore not determined by overall skill alone.

The differences are also poorly described by a simple distinction between AI and physical models. AIFS exhibits deficits in the wind and temperature tails, and these persist for AIFS ENS despite its improved performance in the centre of the distribution. By contrast, the ensemble means of EPT-2 HRRR and EPT-2.1 Europa remain competitive at gale force and in heat. This cannot be explained by ensemble-mean evaluation alone: the ECMWF ENS mean also loses skill in the heat regime. The largest heat deficit belongs to the physical GFS model, while the physical ICON-EU is comparable with the leading regional AI models in the wind and temperature tails. For precipitation, EPT-2 HRRR is the only regional model with positive skill from light through high rain.

These comparisons do not identify the cause of the observed differences. The models vary simultaneously in architecture, training objective, resolution, training data, initialisation, forecast cadence, and spatial domain (Bonavita, 2024; Subich et al., 2025). Performance cannot therefore be attributed to a single distinction such as AI versus physics, regression versus generative training, or global versus regional resolution. Moreover, evaluating ensembles through their means measures point-forecast accuracy but provides no evidence about calibration, spread, or extreme-event probabilities.

The shared conditional-error pattern requires similar caution. Because regimes are defined from the verifying observation, overprediction at low values and underprediction at high values can arise from limited predictability, unresolved local variability, representativeness and observation errors, or regression towards the conditional mean. It should not be interpreted as direct evidence of model smoothing. Nevertheless, the pattern is relevant because it coexists with clear differences in relative MAE. At the heaviest precipitation intensities, for example, EPT-2 Reasoning remains slightly ahead of IFS, EPT-2.1 Europa is close to it, and EPT-2 HRRR, EPT-2e, and GFS fall below it.

Absolute attenuation and relative skill are therefore complementary properties. The evaluated AI models can underestimate extreme observations without being uniformly less accurate than physical models, while strong average performance does not guarantee an advantage in every tail. Within this evaluation, performance at extremes is best understood as a property of individual models and variables rather than of AI weather forecasting as a single class.

6 LIMITATIONS

The regime thresholds are estimated from ERA5 rather than from station records. An “extreme” therefore denotes a value that is unusual relative to the local ERA5 1991–2020 climatology, conditioned on calendar day and UTC hour; it does not necessarily represent an impact-defined heatwave, gale, or record event. Because ERA5 represents grid-cell averages, it has lower variance than point observations. The wind and solar tails consequently contain approximately 10% of observations each, making them less selective than the corresponding temperature tails.

Precipitation is evaluated against a separate rain-gauge panel covering eleven countries, with Italy and Poland absent. Its thresholds are derived from the ERA5 wet-hour distribution and therefore retain the same mismatch between gridded climatology and point observations. Hourly gauge accumulations are also noisier than the synoptic wind and temperature measurements. The heavy-

precipitation results should consequently be interpreted as a relative comparison with IFS on this station panel, not as an absolute assessment of convective-scale forecast skill.

Although the ten-month evaluation contains many forecast–observation pairs, it does not span a complete annual cycle and samples a limited set of meteorological events. It supports the climatological regime comparisons reported here but cannot characterise every form of rare, persistent, compound, or record-breaking weather.

Ensemble models are evaluated through their means. The study therefore measures point-forecast accuracy and does not assess calibration, spread, event probabilities, or other aspects of probabilistic forecast quality. Nor can differences between regression, generative, and physical models be attributed to their training objectives: the models also differ in resolution, inputs, architecture, initialisation, domain, and forecast cadence.

The reported results use a common causal bias-correction procedure estimated from the preceding four weeks of station errors. They compare models after operational post-processing rather than entirely raw forecasts. In addition, ECMWF ENS contributes to the precipitation comparison only at its native 6-hourly leads, whereas the other precipitation models are evaluated hourly.

7 CONCLUSION

AI weather models do not exhibit a uniform MAE deficit relative to ECMWF IFS in the observed-value tails considered here. In Track A, EPT-2.1 Europa leads for wind, EPT-2 HRRR for temperature, and EPT-2.1 Helios for solar radiation. The largest precipitation gains occur at moderate intensities and diminish towards the heavy tail.

The counterexamples are equally important. AIFS and AIFS ENS remain below IFS in the wind and heat tails, GFS has the largest heat deficit, and EPT-2.1 Helios loses skill in the centre of the solar distribution. Track B yields the same model-dependent picture: Europa and ICON-EU are comparable at gale force and in heat, while EPT-2 HRRR performs strongly across light to high rain but not at the heaviest intensities.

Every evaluated model nevertheless overpredicts low observations and underpredicts high observations after bias correction. Models can therefore attenuate extreme values without being uniformly less accurate than IFS within those regimes. In this evaluation, performance at extremes is a property of individual models and variables, not a general distinction between AI and physical weather forecasting.

ETHICS STATEMENT

Several of the forecast systems evaluated in this study are developed by the authors’ organisation. The verification protocol is applied identically across AI and physical systems, every comparison is reported against the independent ECMWF IFS reference, and the archived requests, aggregates and reproduction code described below are provided to support independent scrutiny of the comparisons. The study uses forecast output and meteorological station observations only; it involves no human subjects or personal data, and no station identities or per-station series are released.

REPRODUCIBILITY STATEMENT

Metrics were computed through a content-addressed station-verification API; requests and responses are archived. The reproduction package includes the aggregates used for the reported numbers, headline skill tables, and the extraction, aggregation and figure code. All score-based figures and tables regenerate offline from those aggregates. The package is provided as anonymised supplementary material.

AI USE STATEMENT

Generative AI tools were used to assist with software changes in the extraction, aggregation and figure code, to inspect generated results, and to draft and edit manuscript text. They were not used to generate or modify observations or forecast data. The authors checked AI-assisted code through

the pipeline’s executable self-tests and regenerated artifacts, verified the quoted numbers against the aggregate data, reviewed all manuscript text, and take full responsibility for the content.

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A SUPPLEMENTARY MATERIAL

Table S1: Track A MAE skill (%), 6–48 h, by regime – Figure 5, wind. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.4 (0.5)	-1.7 (1.4)	+6.4 (0.9)	+11.3 (0.7)	+8.5 (0.4)	+9.0 (1.2)
Jua EPT-2 HRRR	+4.5 (0.3)	+2.7 (1.0)	+1.4 (0.9)	+5.1 (0.3)	+4.9 (1.0)	+6.1 (0.9)
Jua EPT-2 Reasoning	+3.0 (0.1)	-0.7 (0.4)	+2.8 (0.5)	+5.6 (0.3)	+4.5 (0.3)	-0.7 (0.3)
Jua EPT-2e	-0.0 (0.2)	-9.1 (1.0)	-1.4 (1.1)	+5.6 (0.5)	+2.9 (0.6)	-6.6 (0.5)
Microsoft Aurora	-0.7 (0.3)	-6.6 (1.1)	-3.9 (1.1)	+1.9 (0.8)	+2.6 (0.8)	-3.4 (0.6)
ECMWF AIFS	-5.4 (0.4)	-9.8 (1.2)	-8.0 (1.3)	-2.1 (0.6)	-1.4 (0.8)	-10.6 (0.6)
ECMWF AIFS ENS (mean)	-2.3 (0.5)	-12.0 (1.2)	-3.3 (1.5)	+3.8 (0.7)	-0.1 (0.7)	-9.0 (0.7)
ECMWF ENS (mean)	+1.4 (0.2)	-0.9 (0.7)	+1.5 (0.6)	+3.1 (0.4)	+2.1 (0.4)	-1.0 (0.2)
NOAA GFS	-11.7 (0.7)	+1.8 (1.4)	-9.5 (1.4)	-21.8 (1.4)	-16.3 (1.3)	+0.6 (0.7)
DWD ICON Global	+3.6 (0.4)	-4.1 (1.2)	+2.8 (1.0)	+2.6 (0.7)	+2.5 (0.7)	+9.5 (1.1)

Table S2: Track A MAE skill (%), 6–48 h – Figure 5, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+11.5 (0.7)	+4.7 (1.0)	+10.8 (1.3)	+11.0 (1.0)	+11.8 (1.1)	+15.0 (1.7)
Jua EPT-2 HRRR	+12.1 (0.4)	+9.5 (2.4)	+11.5 (1.2)	+9.4 (0.9)	+11.4 (1.0)	+19.6 (2.2)
Jua EPT-2 Reasoning	+4.6 (0.4)	-6.9 (1.3)	+2.6 (1.3)	+7.4 (0.4)	+7.0 (0.8)	+1.1 (0.7)
Jua EPT-2e	+0.8 (0.7)	-24.3 (3.6)	-5.0 (2.8)	+6.1 (0.8)	+7.4 (1.1)	-5.0 (0.7)
Microsoft Aurora	-1.3 (1.1)	-22.2 (4.4)	-7.0 (4.6)	+4.1 (1.0)	+3.8 (1.6)	-6.9 (1.0)
ECMWF AIFS	-1.4 (0.6)	-27.3 (3.6)	-5.1 (2.3)	+2.3 (1.0)	+4.9 (1.4)	-4.9 (2.0)
ECMWF AIFS ENS (mean)	+0.2 (0.6)	-24.9 (3.2)	-3.1 (2.0)	+4.5 (0.9)	+6.2 (0.9)	-5.1 (1.7)
ECMWF ENS (mean)	-1.7 (0.3)	-12.2 (1.0)	-3.3 (0.9)	+1.4 (0.4)	+0.3 (0.8)	-6.1 (0.5)
NOAA GFS	-23.9 (1.4)	-25.3 (3.5)	-26.3 (2.9)	-26.5 (2.5)	-17.8 (2.1)	-22.8 (2.0)
DWD ICON Global	+5.5 (0.9)	-3.8 (1.2)	+3.0 (1.7)	+4.1 (1.5)	+8.3 (0.9)	+10.2 (2.0)

Table S3: Track A MAE skill (%), 1–48 h hourly – Figure 5, solar. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+10.2 (1.7)	+15.4 (5.0)	+16.4 (3.4)	-8.2 (1.1)	+13.7 (2.9)	+24.8 (5.4)
Jua EPT-2.1 Europa	+7.1 (1.0)	+3.1 (4.0)	+11.0 (3.1)	+9.2 (2.3)	+1.9 (4.3)	+4.4 (4.8)
Jua EPT-2 HRRR	+0.6 (1.4)	-3.6 (4.6)	+3.8 (3.7)	+4.5 (1.0)	-4.6 (4.8)	-4.1 (4.9)
Jua EPT-2 Reasoning	+6.3 (0.5)	+0.9 (1.3)	+9.5 (0.8)	+14.0 (0.7)	-0.8 (1.9)	-2.6 (2.6)
Jua EPT-2e	+2.0 (1.4)	-10.6 (3.3)	+5.9 (2.3)	+13.5 (1.6)	-7.5 (4.1)	-7.6 (6.0)
NOAA GFS	-11.2 (1.6)	-3.3 (4.4)	-24.3 (2.1)	-28.3 (2.5)	+8.8 (3.4)	+16.7 (4.2)
DWD ICON Global	+5.6 (0.8)	+6.5 (3.3)	+8.1 (2.6)	+3.1 (1.5)	+4.2 (3.3)	+6.9 (4.2)

Table S4: Track A MAE skill (%), 1–12 h – Figure 6, wind.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.0 (0.5)	-3.7 (1.0)	+3.6 (1.1)	+10.5 (0.7)	+8.1 (0.3)	+8.2 (0.9)
Jua EPT-2 HRRR	+2.0 (0.3)	+0.7 (1.0)	-2.1 (0.8)	+2.4 (0.2)	+3.0 (0.9)	+4.7 (0.9)
Jua EPT-2 Reasoning	+1.5 (0.1)	-1.5 (0.2)	+0.4 (0.3)	+3.2 (0.2)	+3.4 (0.2)	-0.4 (0.2)
Jua EPT-2e	-2.4 (0.2)	-10.8 (0.7)	-6.0 (0.7)	+2.1 (0.5)	+1.7 (0.4)	-6.2 (0.4)
ECMWF ENS (mean)	+0.4 (0.3)	–	–	–	–	–
NOAA GFS	-10.9 (0.8)	+4.8 (0.8)	-5.6 (0.9)	-21.5 (1.8)	-18.1 (1.4)	+0.7 (0.6)
DWD ICON Global	+3.7 (0.3)	-1.3 (0.6)	+4.0 (0.9)	+3.3 (0.7)	+1.5 (0.4)	+8.1 (1.1)

Table S5: Track A MAE skill (%), 1–12 h – Figure 6, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.3 (0.8)	+2.9 (1.1)	+7.3 (1.0)	+7.8 (1.1)	+7.5 (1.1)	+13.3 (1.4)
Jua EPT-2 HRRR	+9.5 (0.6)	+9.0 (2.3)	+8.7 (1.6)	+6.2 (1.0)	+8.2 (0.8)	+18.8 (2.0)
Jua EPT-2 Reasoning	+2.4 (0.2)	-5.3 (0.6)	+1.0 (0.6)	+4.3 (0.2)	+4.2 (0.5)	+0.0 (0.5)
Jua EPT-2e	-2.5 (0.6)	-24.3 (2.7)	-7.7 (2.2)	+2.3 (0.7)	+3.0 (1.1)	-7.2 (0.5)
ECMWF ENS (mean)	-3.0 (0.2)	–	–	–	–	–
NOAA GFS	-26.9 (1.5)	-29.6 (4.0)	-32.4 (3.5)	-29.0 (2.4)	-20.5 (1.6)	-23.9 (2.5)
DWD ICON Global	+5.3 (0.8)	-2.9 (1.0)	+2.8 (1.2)	+4.9 (1.5)	+6.3 (0.7)	+10.1 (1.4)

Table S6: Track A MAE skill (%), 1–12 h – Figure 6, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+13.5 (1.9)	+20.6 (5.9)	+20.1 (2.9)	-6.9 (1.5)	+17.7 (2.6)	+28.7 (6.1)
Jua EPT-2.1 Europa	+6.5 (1.4)	+1.2 (3.9)	+10.5 (3.8)	+8.1 (3.0)	+1.9 (4.3)	+5.6 (4.8)
Jua EPT-2 HRRR	-2.1 (1.8)	-6.8 (4.7)	+0.9 (4.3)	+0.3 (1.5)	-6.4 (5.0)	-3.7 (4.8)
Jua EPT-2 Reasoning	+4.4 (0.4)	+0.6 (1.3)	+7.4 (0.8)	+11.0 (0.7)	-2.4 (1.3)	-3.0 (1.7)
Jua EPT-2e	-1.1 (1.7)	-13.5 (3.4)	+4.1 (3.1)	+11.0 (1.7)	-13.8 (4.5)	-10.0 (5.0)
NOAA GFS	-12.7 (1.9)	-5.8 (4.2)	-27.6 (2.7)	-29.8 (2.6)	+9.7 (4.2)	+15.9 (3.6)
DWD ICON Global	+6.4 (1.2)	+6.2 (3.4)	+9.7 (3.0)	+3.4 (1.6)	+4.1 (2.9)	+8.8 (4.8)

Table S7: Track A precipitation MAE skill (%), 1–12 h (Figure 8). Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+1.9 (± 0.5)	-2.9 (± 1.0)	+3.3 (± 1.0)	+8.2 (± 1.0)	+7.8 (± 0.7)	+2.2 (± 0.5)
Jua EPT-2 HRRR	-2.7 (± 1.1)	-10.9 (± 3.8)	+3.9 (± 1.2)	+10.8 (± 1.5)	+6.8 (± 1.1)	-4.5 (± 0.8)
Jua EPT-2e	+4.6 (± 0.9)	+12.4 (± 3.4)	+8.9 (± 1.1)	+11.2 (± 1.1)	+2.2 (± 1.5)	-6.0 (± 1.3)
Jua EPT-2.1 Europa	+2.2 (± 1.5)	+22.0 (± 3.2)	-16.1 (± 4.2)	-16.5 (± 4.6)	-10.8 (± 3.2)	-1.1 (± 0.7)
NOAA GFS	-7.3 (± 1.0)	-3.4 (± 3.6)	-9.1 (± 1.4)	-14.2 (± 1.5)	-16.1 (± 1.9)	-4.8 (± 1.8)
DWD ICON Global	-1.4 (± 2.3)	+11.4 (± 3.3)	-17.4 (± 3.8)	-17.1 (± 4.5)	-11.5 (± 3.5)	-0.3 (± 0.6)

Table S8: Track A MAE skill (%) by lead – Figure 1, wind. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.3	+8.8	+8.7	+9.3	+8.5	+9.2	+8.1	+9.0
EPT-2 HRRR	+3.3	+5.0	+3.9	+5.6	+4.6	+6.0	+5.1	+6.9
EPT-2 Reasoning	+1.9	-0.7	+2.3	-0.8	+2.9	-0.9	+3.9	-0.4
EPT-2e	-1.2	-8.0	-1.0	-7.5	-0.1	-6.7	+1.1	-5.5
Aurora	-1.6	-4.0	-1.6	-3.6	-0.9	-3.6	+0.4	-2.8
AIFS	-6.6	-10.8	-6.5	-11.0	-5.7	-11.1	-4.0	-9.5
AIFS ENS	-3.5	-8.3	-3.4	-8.9	-2.7	-9.4	-0.7	-8.9
ENS	+0.3	-0.9	+0.7	-0.8	+1.2	-0.9	+2.3	-1.5
GFS	-11.1	+2.0	-11.1	+1.8	-11.8	+0.6	-12.3	-0.5
ICON Global	+4.6	+10.1	+4.6	+10.1	+4.0	+9.6	+2.4	+8.6

Table S9: Track A MAE skill (%) by lead – Figure 2, temperature. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.4	+12.5	+12.0	+15.8	+11.5	+14.5	+11.6	+14.4
EPT-2 HRRR	+10.1	+19.5	+12.6	+21.3	+12.1	+19.4	+12.2	+18.4
EPT-2 Reasoning	+2.2	-0.5	+3.4	+0.2	+4.5	+0.9	+6.0	+2.2
EPT-2e	-4.2	-10.0	-1.5	-7.5	+1.1	-4.7	+3.2	-2.7
Aurora	-4.6	-7.8	-2.3	-7.1	-1.2	-7.0	+0.2	-6.7
AIFS	-4.5	-7.5	-2.5	-5.2	-1.7	-5.0	+0.4	-3.7
AIFS ENS	-4.3	-10.1	-1.3	-6.6	+0.1	-5.0	+2.2	-3.3
ENS	-3.0	-6.7	-2.6	-6.3	-1.9	-6.0	-0.6	-5.6
GFS	-27.3	-24.6	-24.2	-22.5	-24.2	-23.2	-22.3	-22.6
ICON Global	+5.1	+10.1	+6.9	+11.7	+6.0	+10.2	+4.0	+8.8

Table S10: Track A MAE skill (%) by lead – Figure 3, solar (Ovc. = overcast, P5–25). Point estimates; SEs in regime tables.

Model	1 h		6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+26.1	–	+10.9	+19.0	+9.0	+15.2	+8.1	+15.4	+9.2	+15.5
EPT-2.1 Europa	+6.2	–	+5.6	+10.1	+6.7	+10.0	+6.1	+10.2	+6.3	+10.2
ICON Global	+7.1	–	+5.6	+9.8	+6.2	+8.9	+5.2	+8.1	+4.3	+6.3
EPT-2 Reasoning	+3.5	–	+4.2	+7.0	+5.1	+8.5	+6.0	+9.5	+7.8	+11.1
EPT-2e	+0.8	–	-4.5	+2.4	-2.5	+2.2	+0.2	+2.9	+4.1	+8.1
EPT-2 HRRR	-4.6	–	-4.8	-0.5	-2.7	+1.3	-1.7	+2.6	+1.1	+5.1
GFS	-12.8	–	-13.7	-30.2	-12.4	-29.4	-11.7	-25.8	-11.1	-24.3

Table S11: Track B wind MAE skill (%), 1–48 h, by regime (Figure 11).

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.2 (0.4)	+0.1 (0.8)	+3.0 (0.3)	+8.9 (0.6)	+8.2 (0.3)	+10.5 (1.4)
Jua EPT-2 HRRR	+2.4 (0.6)	+4.9 (1.0)	-1.8 (1.7)	+3.5 (0.4)	+1.8 (0.4)	+4.1 (1.5)
DWD ICON-EU	+5.3 (0.5)	-0.7 (1.0)	+1.8 (0.6)	+5.6 (0.5)	+3.9 (0.2)	+11.5 (1.6)

Table S12: Track B MAE skill (%), 1–48 h – Figure 11, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+13.2 (0.6)	+5.9 (1.8)	+11.7 (2.0)	+12.3 (1.3)	+13.5 (0.6)	+16.6 (2.3)
Jua EPT-2 HRRR	+12.0 (0.7)	+12.8 (3.3)	+14.9 (1.4)	+9.2 (0.4)	+8.7 (1.3)	+17.2 (3.5)
DWD ICON-EU	+12.4 (0.8)	+7.7 (1.1)	+11.8 (2.3)	+10.6 (1.1)	+11.3 (1.5)	+17.2 (3.2)

Table S13: Track B MAE skill (%), 1–48 h – Figure 11, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+12.6 (2.5)	+21.6 (4.6)	+18.1 (5.5)	-11.3 (0.5)	+13.5 (5.8)	+32.2 (4.9)
Jua EPT-2.1 Europa	+7.8 (2.0)	+7.4 (3.9)	+12.8 (3.1)	+10.7 (1.7)	+0.1 (4.5)	+3.4 (7.8)
Jua EPT-2 HRRR	+0.8 (2.2)	-0.2 (5.1)	+4.8 (5.7)	+6.5 (0.3)	-6.1 (8.2)	-6.4 (7.7)
DWD ICON-EU	+7.2 (1.4)	+12.4 (3.3)	+12.4 (2.5)	+1.0 (2.2)	+0.9 (3.3)	+10.5 (5.7)

Table S14: Track B MAE skill (%) by lead – Figure 13, wind.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.6	+11.1	+9.3	+12.0	+8.9	+11.9	+8.4	+12.3
EPT-2 HRRR	+2.3	+3.8	+3.3	+5.0	+3.9	+5.4	+4.7	+6.7
ICON-EU	+7.5	+14.6	+7.7	+14.7	+7.1	+13.9	+5.4	+13.7

Table S15: Track B MAE skill (%) by lead – Figure 13, temperature.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+10.7	+12.7	+14.6	+17.0	+13.2	+15.1	+13.5	+16.1
EPT-2 HRRR	+12.0	+17.1	+14.2	+19.3	+12.4	+16.8	+12.2	+16.9
ICON-EU	+13.7	+18.4	+14.9	+19.9	+13.2	+17.2	+10.6	+15.5

Table S16: Track B MAE skill (%) by lead – Figure 13, solar (Ovc. = overcast, P5–25).

Model	6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+12.9	+20.3	+11.4	+17.1	+10.4	+16.4	+11.2	+16.8
EPT-2.1 Europa	+7.2	+13.6	+8.0	+13.5	+6.8	+12.3	+6.9	+11.9
EPT-2 HRRR	-3.5	+3.9	-1.5	+4.7	-1.3	+4.6	+1.1	+6.4
ICON-EU	+7.4	+14.1	+7.8	+13.4	+6.0	+12.0	+4.4	+9.3

Table S17: Track B precipitation MAE skill (%) by lead (Figure 15). Point estimates.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
Jua EPT-2.1 Europa	+1.9	+0.5	+9.9	+1.0	+9.0	+1.6	+7.7	+1.9
Jua EPT-2 HRRR	-2.5	-0.6	-2.6	+0.6	-0.9	+1.0	+0.4	+1.7
DWD ICON-EU	-0.1	+0.4	+8.7	-0.6	+7.1	-0.1	+3.2	-1.2

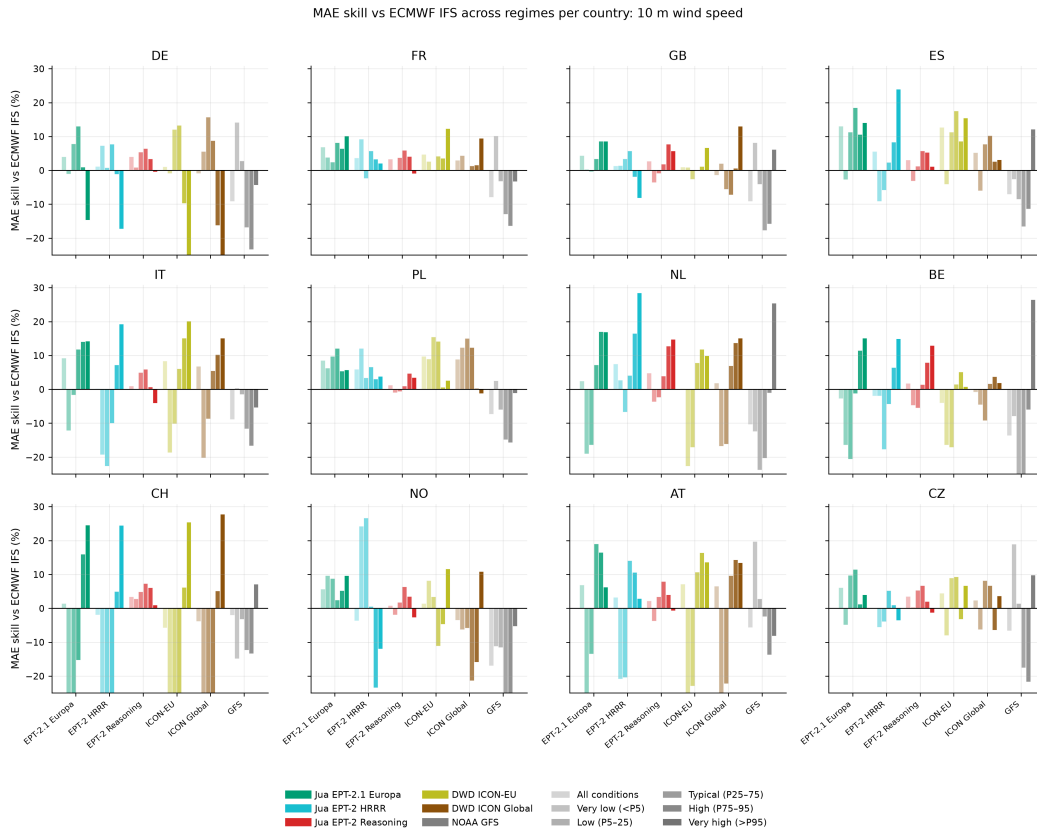


Figure S1: Per-country MAE skill by ERA5 1991–2020 climatological regime, 10 m wind, March–June 2026, pooled 1 to 48 h (hourly). Mixed regional/global model set (Section 4.5).

Table S18: Track B MAE skill (%) for 1 h precipitation by wet-hour regime, 1 to 48 h, debiased – Figure 14. Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.8 (± 0.9)	+29.3 (± 1.4)	-19.1 (± 6.0)	-20.2 (± 7.2)	-9.0 (± 5.7)	+1.2 (± 0.8)
Jua EPT-2 HRRR	+1.4 (± 3.4)	-3.6 (± 9.7)	+10.3 (± 4.4)	+15.6 (± 6.1)	+10.4 (± 2.5)	-0.7 (± 0.3)
DWD ICON-EU	+3.5 (± 0.5)	+15.3 (± 0.3)	-10.9 (± 2.7)	-11.4 (± 3.8)	-6.9 (± 2.2)	+0.1 (± 0.5)

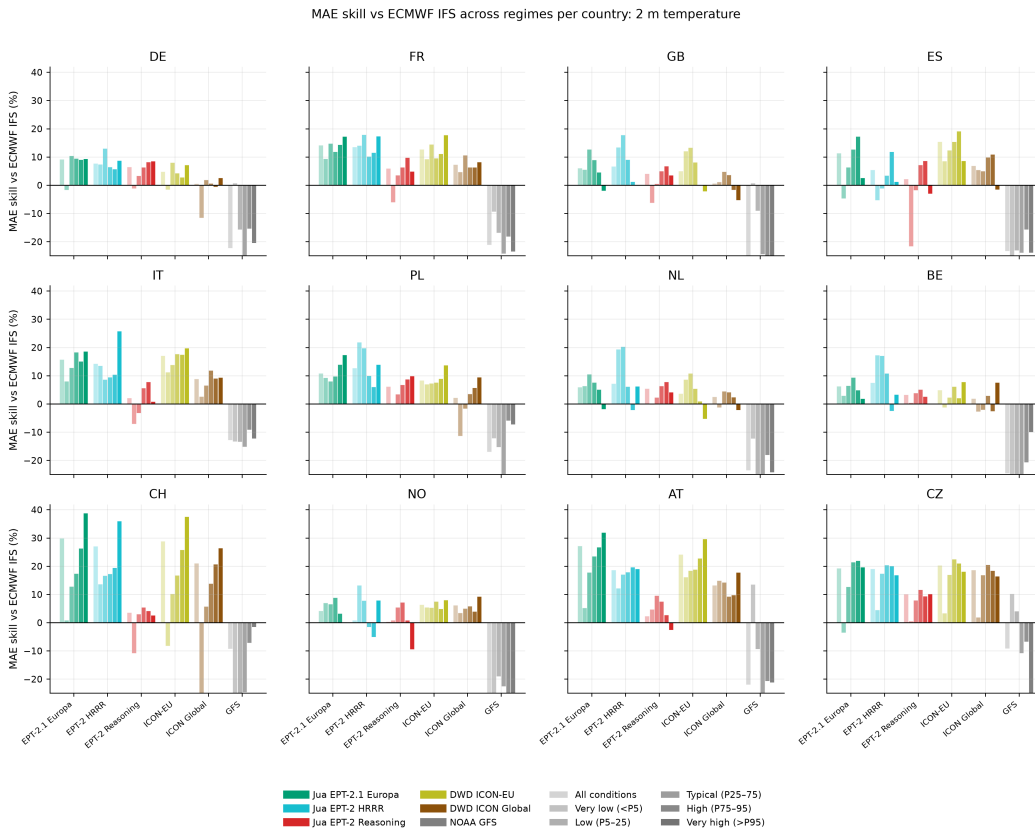


Figure S2: As Figure S1, for 2 m temperature.

Table S19: Heavy-precipitation (>P95) detection on Track A, 1 to 48 h, debiased – Figure 9. POD: probability of detection; FAR: false-alarm ratio; CSI: critical success index; frequency bias is forecast over observed event count.

Model	POD	FAR	CSI	Freq. bias
Jua EPT-2.1 Europa	0.28	0.62	0.19	0.74
DWD ICON Global	0.29	0.65	0.19	0.82
Jua EPT-2 Reasoning	0.23	0.56	0.18	0.53
ECMWF IFS	0.25	0.65	0.17	0.71
NOAA GFS	0.25	0.66	0.17	0.74
Jua EPT-2 HRRR	0.20	0.53	0.16	0.42
Jua EPT-2e	0.20	0.55	0.16	0.44
ECMWF ENS (mean)	0.15	0.50	0.13	0.29

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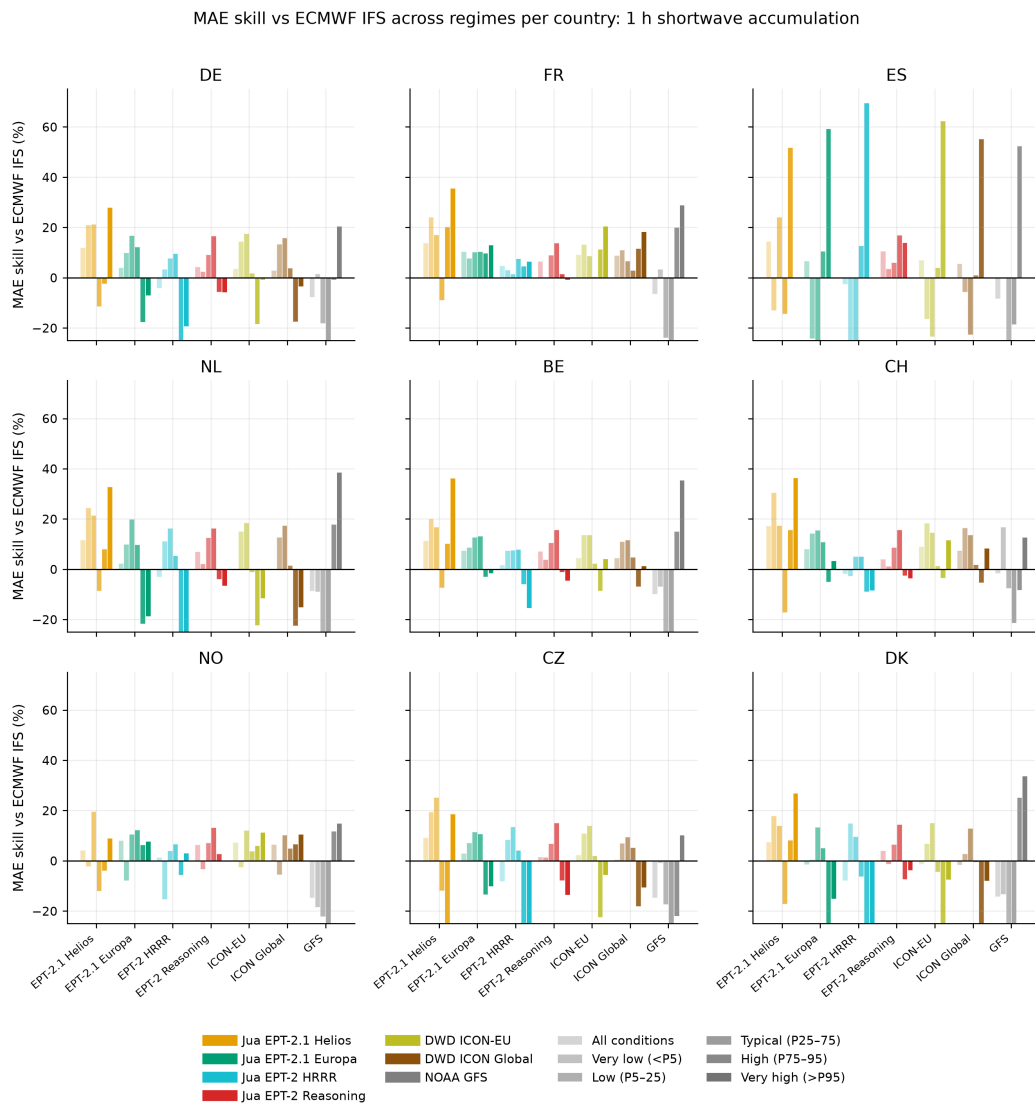


Figure S3: As Figure S1, for 1 h shortwave, with EPT-2.1 Helios added. GB, IT and PL have no solar station observations (Section 3.2).

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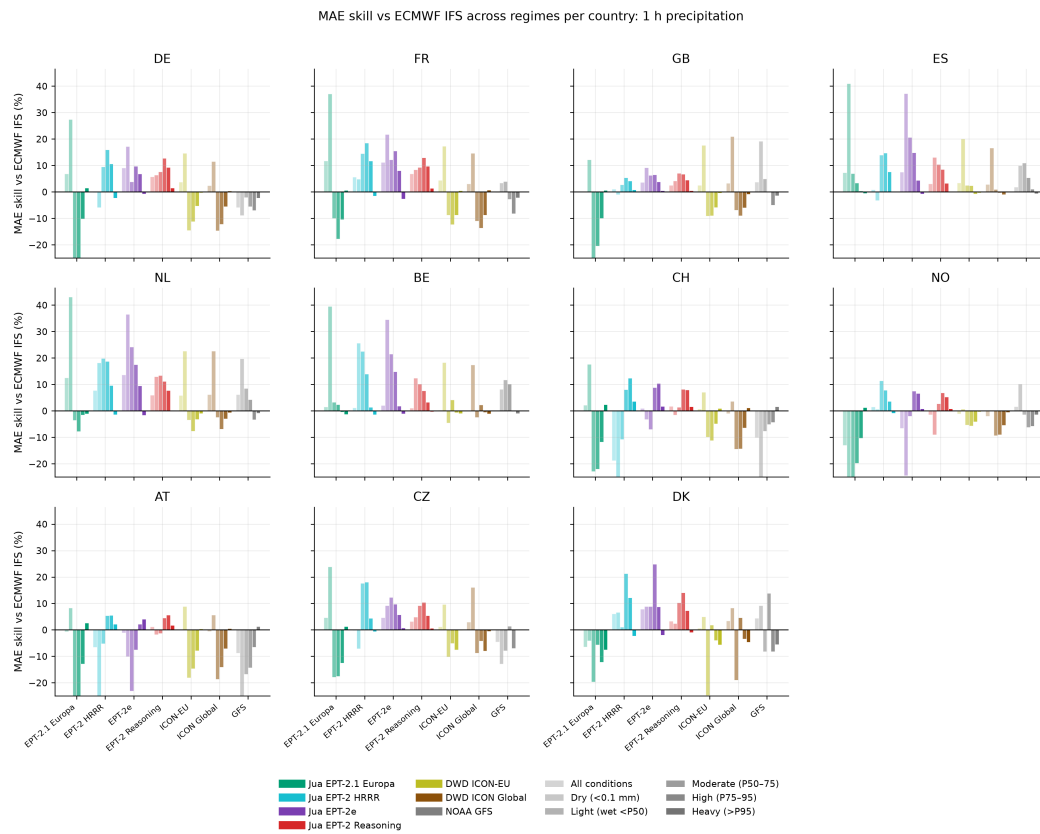


Figure S4: As Figure S1, for 1 h precipitation using the wet-hour regimes of Table 3. IT and PL have no gauges in the precipitation panel.